

Executive Mobility in the United States, 1920–2023^{*}

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Abstract

This paper studies the evolution of U.S. public-firm executive mobility from 1920 to 2023. Executive mobility exhibits a secular increase from the 1920s through the turn of the century. After peaking in the late-1990s, it declined sharply and stabilized at a lower level: chief executive mobility in the 2010s was less than half its late-1990s peak, returning to levels last seen in the 1960s and 1970s. Finance-chief mobility shows a stronger secular rise over the century. We argue that changes in the size of the executive labor market, the importance of general managerial skills, and the redeployability of capital help explain these trends.

Keywords: Executive labor market; Mobility; Reallocation; Labor market trends

JEL codes: G30, J41

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1. Introduction

A rich literature examines the labor market for corporate executives. This research is important because it sheds light on trends in labor-market conditions, the skill sets required to be a successful executive, and the use and effects of executive compensation. Influential papers by [Huson et al. \(2001\)](#) and [Murphy and Zabojnik \(2007\)](#) document that chief executive officer (CEO) turnover and external-to-the-firm hires increased through the last few decades of the 20th century, which they attribute to general managerial skills becoming more important.¹ Consistent with these implications, the number of firms and occupations in which a typical executive works during her career increased starting in the 1970s ([Frydman, 2019](#)). Other research documents that the level of executive pay and the importance of equity-based compensation increased significantly beginning in the 1980s (e.g., [Murphy, 1999](#); [Gabaix and Landier, 2008](#); [Frydman and Saks, 2010](#)). A common thread in this literature is increasing executive job mobility through about 2000, which is when the data used in many of these papers end.

Our paper adds to the executive labor-market literature in six ways. The first two relate to the fact that the extant literature primarily focuses on the period from 1970 through about 2000. We explore mobility in a broader setting and document new patterns over the past century by constructing a new dataset of executive moves between U.S. public firms over the 1920–2023 period. The dataset contains more than 11,000 executive moves and over 395,000 unique executives, involving nearly 22,000 unique firms.

Our first contribution using this more than century of data is to document long-term trends in executive mobility over the last eight decades of the 20th century. While before 1986 fewer than 5% of departing CEOs became executives of other firms within two years, more than 7% did so during 1986–2000.² Moreover, chief executives moved across an increasingly diverse set of industries over time, and their job moves appear to trade off job title prestige with pay.

Our second contribution is documenting a sharp decline in executive mobility that started

¹See also [Kaplan and Minton \(2012\)](#), [Custódio et al. \(2013\)](#) and [Graham et al. \(2020\)](#) for related evidence.

²Throughout the paper, we separately examine different eras of mobility, motivated by previous research, the historical context of Corporate America, as well as our data, that suggests executive mobility began to increase in the mid-1980s (e.g., [Murphy and Zabojnik, 2007](#)).

in the early 2000s, as executives' propensity to move to new jobs in other firms decreased significantly after 2001.³ By around 2006, CEO mobility had declined to levels not seen since the mid-1980s.

Third, we explore which factors are associated with the long-term evolution of aggregate mobility, and focus particularly on differences in the determinants of executive mobility across different eras of Corporate America. Our four eras are: 1920–1945 (interwar/war mobilization), 1946–1985 (postwar managerial capitalism), 1986–2000 (financial deregulation and the dot-com era), and 2001–2022 (post-dot-com and the global financial crisis). These eras track broad changes in the landscape of Corporate America, and also help frame the historical patterns in executive mobility in our data.

Our analyses suggest several key forces are at play. First, the increasing–then–decreasing trend in executive mobility is associated with the changing size of the public firm executive labor market over the past century. Moreover, the rate of new public firm entries through initial public offerings (IPOs) is strongly correlated with the mobility. These findings suggest that the size of the executive labor market is an important determinant (Moretti and Yi, 2024). Second, measures of the importance of general managerial skills (e.g., the ratio of enrollment in MBA to engineering master's programs and the variety of industries to which executives move) are significantly positively correlated with our measures of aggregate mobility, but *only* before 2000—in the post-2000 era, these proxies for general skills do not track with executive mobility. Third, the executive movements are more frequent when the benefits of reallocation are greater (e.g., M&A activity is high; aggregate or industry ROA dispersion is high).⁴

We elaborate on these findings by linking executive mobility to the “redeployability” of the factors of production. When physical assets and labor are more redeployable, production technologies are likely more similar and the managerial skills required may be more transferable. We use a long-run time series measure for the U.S. economy (1920–2023) tracking the redeployability of assets (Kankanhalli et al., 2025; Kim and Kung, 2017), and we con-

³This trend for executives appears to be consistent with a general decline in labor mobility in the US since 2000. See, e.g., “Fewer Americans Uproot Themselves for a New Job,” The Wall Street Journal, August 20, 2018. Related, Decker et al. (2020) document that rank-and-file employee job reallocation rates have declined since the 1980s in the US economy, and in “high-tech” sectors since the early-2000s.

⁴This positive relation contrasts with patterns of capital and non-executive employee reallocation that are shown to be negatively related with benefits measures (e.g., Eisfeldt and Rampini (2006); Saks and Wozniak (2011)).

struct a complementary measure of cross-industry labor mobility. Both variables exhibit the same rise-then-fall pattern as CEO mobility around the year 2000 and are significant predictors of CEO moves in 1986–2000 (increasing CEO mobility) and 2001–2022 (decreasing CEO mobility). This result suggests a deep connection between production and managerial skill, which our data is uniquely able to point out.

Our fourth contribution is the development of a new measure of executive mobility. This measure is the product of two components: (i) the propensity with which a departing executive finds new executive work rather than retiring from executive work, and (ii) turnover (which creates job openings). Much of the existing literature uses the second component (turnover; e.g., [Huson et al., 2001](#); [Murphy and Zabojnik, 2007](#)) as a proxy for mobility. However, we show that the first component is more important in driving trends in executive mobility, explaining more than two-thirds of the variation in overall mobility.⁵ We also show that while most of the above-mentioned factors explain trends in both components of aggregate mobility, the associations are more consistently significant for the first component.

Fifth, we show for the first time that the mobility of chief financial officers (CFOs) is higher, and increased faster, than that of CEOs during the 20th century.⁶ Combined with the above-mentioned importance of skill generality measures in explaining executive mobility, this comparison of CFO and CEO mobility suggests skill generality as a key determinant of executives' labor-market mobility. The faster-rising CFO mobility since the mid-1980s is consistent with CFOs playing an increasingly active role across broad functional areas over the past several decades (see, e.g., [Groysberg et al., 2011](#)).

Lastly, we study the long-run relations between executive compensation and executive mobility, and provide some stylized facts on the relation of mobility with pay levels, the structure of contracts, the relation between pay and firm size, and dispersion in pay. While we do not make causal claims on these relations, our results suggest a tight link between compen-

⁵Some executive mobility research also uses career paths of given individuals to measure mobility, and often finds a different mobility trend. For example, we document that executive mobility trended up throughout most of the last century. The finding contrasts somewhat with a finding in [Frydman \(2019\)](#) that the mobility of executives at largest public firms, measured by the number of firms a given executive worked for during her career, declined between the 1930s and 1960s.

⁶Between 1920–1949 and 1950–1985, the propensity of departing CFOs to become CFOs of other companies increased from 1.4% to 3.3%. But during 1986–1999, this propensity jumped to 9.7%, triple the propensity of CEO-to-CEO moves during the same period.

sation and mobility across the history of Corporate America, expanding on the work done by [Frydman and Saks \(2010\)](#).

2. Data and Sample Selection

We construct a comprehensive database of corporate officers, such as the chief executive officer (CEO), chief financial officer (CFO), and various corporate vice presidents (VPs), and their movements across US public firms from 1920 to 2023. We combine information from a number of sources. First, we hand-collect names of corporate executives, as well as financial data on their firms, from Moody's Industrial Manuals ('Moody's') from 1920 to 1988, and also the year 1998. Second, we collect names of corporate executives from Compact Disclosure during 1985-2004. Third, we combine data from Execucomp, Equilar and scraped SEC filings to provide a full picture of executives at US public firms in the 2005-2023 period.⁷ We then use news searches to manually check the validity of all cases of CEO and CFO moves, and collect the new job titles of moving executives and whether the moves were associated with mergers and acquisitions or subsidiary relations.

Our full sample includes 248,156 firm-year observations for 22,340 unique firms and 395,114 executives (including 55,370 CEOs and 51,891 CFOs) from 1920 to 2023. We describe trends in executive mobility using this full sample (see the next section), as well as subsamples for fiscal years 1920-1945, 1946-1985, 1986-2000 and 2001-2022.

3. Trends in Executive Mobility in the Past Century

This section examines long-run trends in the mobility of US corporate executives from 1920-2023 using a sample of executives' movements across firms. We provide evidence that executive mobility across firms and industries increased over the past century; in contrast, mobility trends have reversed in the 21st century. We construct formal variables that measure executive mobility across companies.

⁷Though we collect data from the SEC, Equilar and Execucomp through filing year of 2024 (which covers fiscal year of 2023 for most firms), our sample is only a partial snapshot of fiscal year 2023. Thus, we limit most of our analysis to end in FY2022.

3.1. Propensity and Nature of Executive Mobility over the Century

The breadth of the US executive labor market has changed considerably over the past 90 years. The first four tables explore whether these changes led to improved mobility for executives, CEOs and CFOs in particular. We first document an increase in the magnitude of how often an executive leaving one job takes a job at another company, rather than retires from executive work. Table 1, Panel A shows that among the CEOs who left their jobs during 1920-1945 and 1946-1985, only 3.14% ($= 35/1,113$) and 5.14% ($= 271/5,270$) moved to other firms to become executives within the next two years.⁸ In comparison, during 1986-2000, 7.21% ($= 719/9,969$) of former CEOs became officers at other firms. Lastly, in the 2001-2023 era, only 4.6% ($= 875/19,014$) of former CEOs became executives at new firms. This suggests that the patterns in executive mobility observed in the 1986-2000 period are unique in the history of Corporate America.

A more detailed examination reveals that the propensity of former CEOs to become CEOs at new firms increased in the later two eras. 276 and 485 departing CEOs in the 1986-2000 and 2001-2022 eras became CEOs at new firms (2.77% and 2.55% of all CEO turnovers, respectively). The rates of CEO-CEO mobility in the earlier eras are much lower.⁹ The fraction of former CEOs who move to become non-CEO executives at their new firms also increased, from 1.89% during 1920-1945 to 3.97% in the 1946-1985 era, and 4.44% in the 1986-2000 era. This number fell to 2.05% in the 2001-2022 era.

Turning to the mobility of CFOs, Panel B shows that the propensity of CFOs who leave their initial firms to accept executive positions at other firms increased even more than that of CEOs over the past century. During 1920-1945, only 2.71% of departing CFOs moved to become executives at other companies. By 1946-1985, this propensity nearly doubled to 4.87% and increased again to 13.74% over 1986-2000, and stayed relatively high at 10.64% in the latest era. By this period, the average CFO's propensity to move to another firm is considerably

⁸In Table 1 and throughout the paper, we define an executive move as a given executive leaving a firm and becoming an executive at another firm within two years after the departure.

⁹These fractions of CEO-to-CEO moves are similar to estimates in Gibbons and Murphy (1992) that 2.2% of 1,631 departing CEOs in their dataset become CEOs at other firms during 1970-1988. Also, they find that approximately 3.8% of the departing CEOs take a non-CEO position at another firm, similar to our finding that about 4.1% of departing CEOs became non-CEO officers during 1950-1985.

higher than the average CEO's (4.6%) propensity to move. We will show later that CFO mobility sharply increased and then decreased around 2000, akin to the CEO mobility trends, but the decrease in the latest era is less pronounced.

This higher and faster-increasing mobility of CFOs relative to CEOs over the past century is consistent with the CFO's skills being more general and thus more easily redeployable across firms. Moreover, anecdotal evidence suggests that CFO skills became increasingly "CEO-like," which would also increase their appeal to other firms.¹⁰ Another non-mutually-exclusive explanation for the faster-increasing CFO mobility relative to CEO mobility (especially after 1985) is increasing visibility of CFOs due to improved communication technologies. In particular, increased exposure of public firm CFOs in earnings calls and in the media may have given finance chiefs improved opportunities to move to other firms.¹¹

In addition to the likelihood of moving to a new firm, we also explore the breadth of new positions. We find that throughout 1920-2023, common first non-CEO job titles of moving CEOs at the new firm include President (36.4%), VP (17.7%), and COO (8.7%). Thus, as explored in more detail below, it appears not uncommon for former CEOs to move as non-CEO executives of larger (and likely more complex) firms, and sometimes ultimately become CEO of the new firm. In comparison, a relatively higher proportion of CFOs (about three quarters) move to a CFO position in their new companies. A plausible explanation for this difference is that CFOs' skills, while more portable across firms and industries, are specific to corporate finance functions. As a result, it is rare that previous CFOs become chief operating officers (COOs) or divisional managers that require operation-specific skills. It is also rare that CFOs become CEOs (see the last row of Table 1), but over time this proportion has grown.

Table 2 explores a related question on the nature of executive moves: Do executive job movements represent external promotions? Answering this question is important to understand how firms might respond to changing external job opportunities of their own executives,

¹⁰See, e.g., Groysberg et al. (2011) and "[Johnson Controls Hires Company Outsider as Finance Chief](#)," The Wall Street Journal, August 18, 2020. These articles suggest that CFOs are increasingly involved with operation-related roles such as product development and technology, as well as traditional roles such as managing acquisitions.

¹¹Also, the magnitude of the CFO pay increase in the last two decades is smaller than the increase in CEO pay, which may have led finance chiefs to continue to work longer than chief executives. Using Frydman and Saks (2010) and ExecuComp data, we find that the average total CEO (CFO) compensation increased by \$2.4 million (\$1.1 million) over 1986-2023.

which we examine in Section 5. The first three columns show that CEOs on average move to larger firms and receive a large pay increase. Over the full sample period (1920-2022), 62.20% of CEO moves are to larger firms, measured by total assets. A larger fraction of CEOs (70.10%) move to a bigger company when they become non-CEO executives at a new firm. In comparison, the table also shows that CEOs become CEOs at larger firms only half the time (50.51% in the 2001-2022 era, for example). This is perhaps surprising, given the literature on competitive assignment (e.g. [Gabaix and Landier, 2008](#)).

Interestingly, pay increases of CEOs moving to be non-CEOs at larger firms are much larger than those moving to be CEOs. CEO-CEO transitions see a 76.15% increase in pay, where pay is measured as the sum of salary, bonus, stock and options pay (measured in fair value).¹² In contrast, the increase in pay for moves to be non-CEO executives (generally at larger firms) is nearly 200%.

These results are consistent with CEOs moving to other firms to receive either (1) job-title-prestige and higher pay, or (2) less-job-title-prestige but a larger pay increase at a larger company, which may offer increased career opportunities in the future. Thus, the full sample results in Table 2 suggest that the majority of CEO moves can be characterized as external promotions across multiple dimensions, in terms of prestige of the employer, career opportunities, and pay. In terms of trends, while moving to larger firms became more common between 1920-1945 and 1946-1985, particularly for CEO-to-non-CEO moves, this trend has reversed in the latest era particularly.

An interesting fact about pay increases when CEOs and CFOs switch firms is that the average pay increase that executives receive in the latest era (2001-2022) has fallen dramatically, particularly for CFOs. This suggests that large pay packages (outside options) tempted executives to new firms in the 1986-2000 era were an important driver of mobility.

3.2. Cross-Industry Mobility of Executives

The previous section shows that the propensity of departing corporate executives to find new executive work increased over the past century. We now examine trends in the frequency at which executives move to jobs in other industries among the subset of moves, which could

¹²We thank [Frydman and Saks \(2010\)](#) for sharing their pre-1992 data with researchers.

be an important element of the mobility patterns documented in the previous section. Table 3 illustrates trends in cross-industry mobility by presenting the fraction of CEO moves from an “origin” to a “destination” industry. For compactness, this “graphical” analysis relies on one-digit SIC codes but in our empirical analysis of executive retention strategies in Section 5, we use two-digit SIC codes. Panel A (1920-1945) and Panel B (1946-1985) show that before 1986, over 60% of CEO moves-to-new-firms, including moves to same- and different-industry firms, occur within the manufacturing sector (SIC = 2 or 3). This magnitude fell to about 50% in the third era (1986-2000), and fell to about 40% in the final era. Thus, Panel C demonstrates a more varied set of origin and destination industries for external CEO hires in recent decades.

We formally examine the diversity of industries across which CEOs move using the Herfindahl-Hirschman Index (HHI) of the fraction of moves between industry pairs. This concentration measure of between-industry moves was 17.55% during 1920-1945, 15.75% during 1946-1985, and decreased to 6.64% and 6.18% during 1986-2000 and 2001-2022, respectively, indicating that CEO moves in the later period are more widely dispersed across industries. In addition, we find that “off-diagonal” movements (i.e., across different industries) became more frequent over the century: the fraction of across-different-industry movements was about 37% during 1920-1945, and increased to 47% and 48% in the second and third eras. The finding that CEOs moved to a more diverse set of industries in recent decades is consistent with the increasing importance of general managerial skills, as opposed to industry- or firm-specific skills, in the executive labor market (Murphy and Zabochnik, 2007; Custódio et al., 2013; Frydman, 2019). However, this cross-industry mobility reversed in the final era, with only 42% of moves, suggesting that perhaps the demand for CEOs’ general managerial skills may have changed somewhat.

Table 4 presents the analogous fraction of CFO moves from one industry to another, and shows that the patterns are largely similar. CFO cross-industry mobility is on average higher in all eras than CEO mobility, consistent with CFO skills applying more readily across industries.

3.3. Trends in Formal Measures of Executive Mobility

3.3.1. Measures of Executive Mobility

To further examine the time-series changes in executive mobility and the factors that drive the trends, we formally construct aggregate measures of executive mobility. Specifically, we measure executive mobility as the total number of CEOs who move to other firms as executives, scaled by the lagged total number of CEOs, or separately, scaled by the lagged total number of CEO turnover events. These aggregate measures capture the propensity at which sitting or leaving CEOs are hired by other firms. (We repeat the same calculations for CFOs.) The resulting measures are defined as follows:

$$Mobility_t = \underbrace{\frac{\# \text{ CEO (or CFO) moves}_t}{\# \text{ CEOs (or CFOs)}_{t-1}}}_{(A)} \quad \text{or} \quad \underbrace{\frac{\# \text{ CEO (or CFO) moves}_t}{\# \text{ CEO (or CFO) turnovers}_{t-1}}}_{(B)} \quad (1)$$

where $Mobility_t$ is an aggregate measure of mobility for CEOs in year t , $\# \text{ CEO moves}_t$ is the number of CEOs in year $t-2$ or $t-1$ who become officers at another firm in year t ; $\# \text{ CEOs}_{t-1}$ is the number of CEOs, and $\# \text{ CEO turnovers}_{t-1}$ is the number of CEO turnovers in year $t-1$. Analogous mobility measures can be constructed for CFOs.

An executive may move to another firm because she is looking for an opportunity at unrelated firms or because of mergers and acquisitions (M&As). To account of the effects of M&As, we construct two versions of our executive mobility measures — the first one accounts for all moves, and the second one excludes moves directly tied to M&As or subsidiary relations (see [Frydman \(2019\)](#) for a related discussion).

3.3.2. Trends in Executive Mobility

Figure 1, Panel A shows that the two measures of CEO mobility that are scaled by the number of CEOs (A in Eq. 1), whether including or excluding M&A / subsidiary-related events, move in parallel from 1921 through 2022 ($\rho = 0.957$).¹³ The dashed gray line showing the difference between the two measures indicates that the portion of mobility due to M&As or across sub-

¹³Our mobility measures require the lagged number of CEOs or turnovers, and thus begin in 1921. It ends in 2022 as we do not have all observations of firms in 2023

sidiaries is relatively stable across the century. That is, M&A trends do not drive aggregate mobility trends. Panel B presents the measures scaled by the number of CEO turnover events (B in Eq. 1), which exhibit similar patterns with the mobility measures shown in Panel A. Given that M&As or subsidiary relations do not drive mobility in an important way, our analysis below relies primarily on the mobility measures that include executive moves associated with M&As or subsidiaries.

Note that our first mobility measure for CEOs in Eq. (1) can be re-written as:

$$\begin{aligned}
 Mobility_t &= \underbrace{\frac{\# CEO\ moves_t}{\# CEOs_{t-1}}}_{(A)} \\
 &= \underbrace{\frac{\# CEO\ moves_t}{\# CEO\ turnovers_{t-1}}}_{(B)} \times \underbrace{\frac{\# CEO\ turnovers_{t-1}}{\# CEOs_{t-1}}}_{(C)} \quad (2)
 \end{aligned}$$

Panel C of Figure 1 presents this decomposition of the first mobility measure (A) into these two components: (B) the number of CEO moves scaled by the number of CEO turnovers (which is the second mobility measure in Eq. (1)), and (C) the turnover rate (i.e., the number of CEO turnovers scaled by the number of CEOs). Thus, our mobility measure (A) nests the two main proxies for executive mobility employed in the literature—the propensity of departing executives to find new executive work (B) and turnover rate (C) (e.g. [Huson et al., 2001](#); [Murphy and Zabojnik, 2007](#); [Frydman, 2019](#)) and allows us to examine which approach is more informative.

Panel C shows that the components reinforce each other when mobility increases and then decreases from the 1980s through 2000s; nonetheless, the first component (B), which captures the propensity of departing executives to find new executive work, is the dominant driver of executive mobility: a statistical decomposition shows that the variation in the log of the first component drives approximately 78% of the variation in the log of the overall mobility measure (and the second component, turnover rate, and its covariance with the first term drive 22%). In addition, the first component is more similar to overall mobility in its steady increase from the mid-1980s to 2000. These results highlight one contribution of our paper relative

to previous work: showing that executive mobility involves departing executives finding new executive-level jobs (B) in addition to executive turnover (C); and documenting that the former is empirically the more important element of mobility.

Both elements of mobility generally trended up throughout most of the sample period, consistent with the overall trends we presented in the previous sections. In an unreported regression, we find that a linear time trend accounts for nearly 30% of variation in aggregate CEO mobility from 1921 to 2023. However, as we explore below, a time-trend does not alone explain trends in executive mobility, given the sharp decline that began in about 2000.

The mobility measures for CFOs shown in Figure 2 exhibit similarly increasing long-run trends until the late-1990s, when they began to decline as did the measures for CEOs. The key difference is that the magnitude of the increase during the late-1980s and 1990s, as well as the peak level in the late-1990s, is much higher for CFOs. As mentioned above, potential explanations for the greater increase in CFOs' mobility relative to CEOs' include the increased visibility of CFOs due to their changing roles (e.g., more like CEOs') and improved communication technology over the last two decades of the 20th century. Nonetheless, CFO mobility exhibits a familiar downward trend post-2000. In the next section, we turn to the analysis of what explains these mobility trends.

4. Explaining Executive Mobility Trends

The previous section documents generally increasing executive mobility in the past century, with the trend reversed in the last two decades of the sample. In this section, we explore potential explanations for these trends, with a focus on explaining the increasing-then-decreasing pattern from the 1980s to the 2020s. Existing research argues and provides some evidence that increasing importance of general managerial human capital can explain the increase in executive mobility during the 1980s and 1990s (e.g. [Murphy and Zabojsnik, 2007](#); [Frydman, 2019](#)). However, this research does not document the subsequent decline in mobility (because the samples used in the previous research end around the turn of the century).

4.1. Labor Market Size and the Benefits of Executive Movements

The past two decades observed a steady decline in the number of publicly-listed firms, reversing a long-run increasing trend before the 2000s (see, e.g., [Doidge et al., 2017](#)). In addition, firm concentration has increased across many industries since the late-1990s ([Grullon et al., 2019](#); [Covarrubias et al., 2019](#)). We begin our investigation by analyzing whether these shifts in market structures are associated with time-series patterns in executive mobility. We initially show graphical evidence for a number of explanatory variables, then present regression evidence in Section [4.3.1](#).

The literature on labor market search and matching shows that larger labor markets facilitate job search, leading to higher labor mobility and match quality (e.g. [Diamond, 1982](#); [Petrongolo and Pissarides, 2006](#); [Moretti and Yi, 2024](#)). Thus, to the extent that the public-firm executive labor market is segmented (vs. that for private firms), changes in the number of public firms would lead to changes in the size of the executive labor market and in turn mobility.¹⁴ Following the literature, we measure the labor market size for public firm executives using the log number of firms on the NYSE, Amex, and Nasdaq (see, e.g., [Petrongolo and Pissarides, 2006](#); [Harmon, 2013](#)).

Panel A of Figure [3](#) shows that the executive labor market size exhibits long-run trends similar to those for mobility, with a correlation of 0.58. As the number of public firms increased continually from the 1920s through the 1990s, our mobility measure also exhibited an increasing trend. Importantly, both the number of public firms and executive mobility experienced a pronounced increasing-then-decreasing pattern since the mid-1980s through the 2000s. This graphical analysis suggests that the changing size of the executive labor markets is a plausible explanation for the key secular mobility trends we document—the long-run increase followed by a sharp decline and stabilization.¹⁵

¹⁴Consistent with segmented labor markets for U.S. public-firm executives, [Cziraki and Jenter \(2022\)](#) find that 80% of executives hired by S&P500 firms are from U.S. public firms.

¹⁵It is important to distinguish the labor market size explanation for declining mobility from a potentially mechanical relation in that our sample is made up of public firms. (So, for example, if the public firms that disappear are each replaced with a private firm, and executives continue to move to these private firms at the same rate as before so “true” mobility has not changed, our measure might mechanically exhibit a decline in mobility). To eliminate this mechanical possibility, in Section [4.6](#), we perform an analysis that also includes private firms and confirm that executive mobility does in fact decline post-2000.

While labor market size is correlated, it is not clear if these job opportunities come from existing public firms or newly-public firms. In Panel B, we analyze the relation between CEO mobility and the proportion of newly-listed to total public firms in the economy. Our analysis is limited to the 1977 and later period, given data availability.¹⁶ The series are strongly correlated ($\rho = 0.65$, which rises to 0.84 if we limit the sample to 1990 and later). As such, newly public firms seem to be an important driver executive mobility, within the entire executive labor market.

In addition to these dynamics with respect to labor market size, the changing nature of labor contracts for executives could explain their mobility trends. [Kini et al. \(2020\)](#) find that the fraction of newly hired CEOs whose contracts are covered by non-compete agreements increased from about 45% in the mid-1990s and about 60% in the early 2000s to close to 70% by the late 2000s. Thus, the increasing prevalence of non-compete clauses for executives, which limit their across-firm mobility, could explain the decline in mobility since the early 2000s.

4.2. Capital Reallocation-Driven Mobility

While the size and thickness of the executive labor market are important drivers, another source of mobility could be reallocation of capital in the economy. Perhaps the most important source of reallocation in the US economy is M&As. This is a distinct source of mobility than the size of the labor market, as it represents the shifting of capital across firms.

Figure 3 Panel C displays the relation between the aggregate number of M&A deals undertaken by public firms and CEO mobility. Importantly, we only consider non-M&A-driven CEO mobility events, to lessen the mechanical correlation between the two series.¹⁷ The two series are highly correlated ($\rho = 0.74$), suggesting that M&A-driven reallocation drives executive reallocation, as the skills of executives are required to manage the factors of production.

¹⁶We thank Jay Ritter for making data available on his website

¹⁷We construct our M&A series from three sources. For 1921-1962, we use the "Historical Statistics of the United States" (1975 edition), which lists FTC-registered M&A deals in the manufacturing, mining and construction industries. For 1963-1982, we use the Mergerstat annual review, which covers all deals with a meaningful transfer of control (e.g., $\geq 10\%$) and a minimum reported value (\$1m deal value). For 1983 and later, we rely on SDC. We note that the samples of firms in these three series may differ.

4.3. General Managerial Skills and Executive Mobility

The previous section shows evidence that changes in labor market size and benefits of executive reallocation are associated with aggregate trends in mobility. We now turn our attention to whether changing importance of general managerial human capital could explain executive mobility trends from the 1960s through 2000s by employing new as well as classical measures. First, the ability of executives to move across industries, as shown in Tables 3 and 4, is intuitively an important component of mobility. We therefore examine whether trends in cross-industry moves is related to variation in executive mobility over time, using the HHI of the fraction of between-industry (one-digit SIC) CEO movements. For this analysis, we calculate the HHI using the past ten years of CEO moves beginning in 1925. Panel D of Figure 3 plots the inverse HHI measure of between-industry movements and our mobility measure, and shows that they are positively correlated ($\rho = 0.58$). A dramatic increase in $1/\text{HHI}$ (which implies a higher degree of cross-industry mobility) since the mid-1980s coincides with a large increase in executive mobility. Also, both series are relatively stable between the 1970s and early-1980s. During the 2000s, the inverse HHI measure plateaus while the mobility measure declines. Hence, propensity of cross-industry moves is a candidate explanation of executive mobility until about 2000 but likely not thereafter.

The second measure of general executive skills is the ratio of new enrollment in MBA degree to engineering master's degree programs in the US, proxied by the number of degrees conferred two years after. We collect data on master's degrees conferred in business and related fields (referred to as "MBA" henceforth) and in engineering and engineering technologies (referred to as "engineering" henceforth) from the US Department of Education, National Center for Education Statistics from 1970 through 2022. (The 1970 start date is due to data availability.) [Murphy and Zabojnik \(2007\)](#) argue that the rising popularity of MBA education during the 1980s-1990s indicates increasing importance of general management skills for corporate executives, relative to firm- or industry-specific skills. Anecdotal evidence suggests however that specialized, perhaps technology-related skills might have become more important for executives in the two most recent decades in our sample given the rise of technology-

oriented firms in which (firm-)specific investments are important (see, e.g., [Gartner, 2019](#)).¹⁸

Panel E of Figure 3 plots both the MBA-to-engineering degree enrollment ratio and our mobility measure from 1970 to 2022. Both variables increased during the 1970s and then decreased until the mid-1980s. Since the late-1980s, both the relative number of MBA program enrollment and the mobility measure increased substantially through the late-1990s. During the 2000s and afterwards, the mobility measure declined substantially, whereas the relative importance of MBA to engineering education experienced a relatively small decrease. Thus, the importance of general managerial skills (over technical skills) is a candidate explanation for executive mobility until about 2000 but likely not thereafter.

Lastly, a particular type of general management skills relates to managing complex organizations by acquiring and restructuring firms and divisions, often across different industries (e.g. [Vancil, 1987](#); [Custódio et al., 2013](#)). In particular, as the market for corporate control emerged in the early 1980s, these general skills were likely more valuable for corporate executives, especially those who oversee large, complex firms.¹⁹ As such, our M&A measure can be interpreted as both a reallocation and general skill-driven measure.

4.3.1. Time-Series Regression Analysis

The previous sections present graphical relations between executive mobility and various economic variables. Some of these variables appear to co-move with the increasing-then-decreasing pattern of executive mobility. In this section, we use a regression framework to formally analyze the time-series relation between variables in Figure 3 and CEO mobility, with a focus on whether the variables help explain the increasing and/or decreasing mobility patterns across eras. Specifically, we estimate the following equation:

$$Mobility_t = \sum_{k=1}^4 \left(\alpha_k \cdot era_t + \beta_k \cdot X_t \cdot era_t \right) + \gamma \cdot M_t + \varepsilon_t \quad (3)$$

¹⁸[Gartner \(2019\)](#), [Top 10 Emerging Skills For the C-Suite](#).

¹⁹See [Murphy \(2013\)](#) for a historical account for the rise of the market for corporate control, and hostile takeovers in particular, Murphy argues that corporate managers' pursuit of inefficient diversification and larger firm size in the previous decades had led to active corporate restructuring markets in the 1980s and 1990s ([Murphy, 2013](#), p. 267).

where era_t represents our four eras: 1920-1945, 1946-1985, 1986-2000, and 2001-2022 and $Mobility_t$ represents our measure of aggregate executive mobility in year t defined in Eq. (1). X_t includes explanatory variables that capture such aggregate factors as executive labor market size, benefits of reallocating executives, and the importance of general managerial skills as introduced in the previous sections. M_t represents a vector of macroeconomic variables including the growth rate in real GDP, BAA-AAA credit spread, aggregate productivity, and three-month T-bill rate, NBER recession indicator, and the aggregate standard deviation of ROA. To account for serial correlations both in the mobility measure and explanatory variables, we use Newey-West standard errors with four years of lag. This specification allows us to study how the drivers of CEO mobility have changed across time, as motivated by Figure 3.

Table 5 summarizes the regression results based on the mobility measure capturing the propensity of sitting executives to find new executive work (A in Eq. 4). In the table, for each column, we display era-specific slopes. For example, column 1 shows the era-specific slope of the relation between CEO mobility and the log number of public firms. We also display the average effect, calculated as the observation-count-weighted effect from the piecewise linear regression.

In column 1, we find that, overall, the size of the labor market drives mobility. However, the relationship is only strongly positive in Eras 3 and 4; a similar finding arises for column 2 when we look at the rate of IPOs in the economy.

In columns 3 and 4, we study the relation between CEO mobility and M&A. On average, there is a strong relation, but the effects are particularly strong in the 1986-2000 era. As such, reallocation overall seems to be an important driver of mobility.

Our two proxies for general skills: cross-industry CEO mobility and MBA entrants into economy both strongly correlate, but only in the 1986-2000 period. In particular, there is a null relation in the 2001-2022 era. This result suggests that general managerial skills, while an important driver in the run-up of executive mobility in 1986-2000, play less of an important role, and perhaps more firm-specific skills dominate (Murphy and Zabojsnik, 2007; Barry et al., 2025).

4.4. Executive Mobility and Mobility of the Factors of Production

The previous analysis suggests that general managerial skills were an important driver of executive mobility in the 1986-2000 period, but the importance has perhaps decreased in the 2001-2022 era. In this section, we explore why this has occurred.

An important aspect of managerial skill is optimally deploying the firm's factors of production. This requires significant understanding of the complex relation between the firm's production process and input factors. It stands to reason then that if factors of production, namely capital and labor, are redeployable *across industries* in the economy, then so too will be executive mobility. As such, we hypothesize that executive mobility should mirror both capital and labor mobility.

To study the relation between capital and executive redeployability across industries, we use an asset redeployability measure from [Kankanhalli et al. \(2025\)](#) and [Kim and Kung \(2017\)](#). This measure, at the aggregate level, measures how well assets used in production are able to be redeployed in different sectors of the economy. [Kankanhalli et al. \(2025\)](#) construct a measure going back to the turn of the 20th century using BEA data, so we employ their measure for the 1920-2023 period.

Figure 4 Panel A displays a visual comparison of executive mobility and asset redeployability. Interestingly, Asset redeployability increases dramatically in the 1980s, and then fell around the year 2000, similar to the trend in executive mobility. Overall, these series correlate strongly ($\rho = 0.59$).

In Panel B, we compare executive mobility with labor redeployability. We measure labor redeployability as the inverse-HHI of workers moving across industries (IND1990 industries, which are akin to SIC3 granularity) from Census Microdata. When this measure is high, workers in the economy move to a broader range of industries when they move. As such, this measure proxies for the redeployability of typical labor in the economy.²⁰ This measure also correlates with executive mobility, and displays the same increase-then-decrease pattern around the year 2000.

²⁰This measure is available from 1976, when the Census started reliably tracking individual-level worker identifiers.

We also undertake time series regression analysis in Table 6. The measures both strongly positively correlate with CEO mobility in the 1986-2000 and 2001-2022 eras, confirming the visual evidence that the fall in redeployability of capital and labor mirrors the fall in executive mobility.

Overall, we interpret these results as evidence that firm-specific, or industry-specific, skill of executives has become more important in later years (Barry et al., 2025; Cziraki and Jenter, 2022). While the cause of this change is not clear from our analysis, the results suggest that a declining redeployability of capital and labor plays an important role.

4.5. Executive Mobility and Compensation

Executive compensation and executive mobility are tightly linked. Our data uniquely positions us to provide descriptive evidence on the relation between the two across the history of Corporate America. In particular, we can study the predictions of important theories on the relation between pay and mobility, and how well these predictions are matched in the data over time. We note that our analysis here is entirely descriptive as the relation between mobility and pay is endogenous.

In Figure 5, we display the time series relation between mobility and different aspects of pay. Panel A displays log total pay. Overall, total pay is not strongly correlated with executive mobility, which may perhaps be surprising given the literature on pay levels and match of CEOs to firms (Gabaix and Landier, 2008). However, we note that the sample composition of this figure has changed over time (Execucomp added more firms above the Frydman and Saks (2010) data).

Existing research shows that equity-based incentive pay could be related to executive mobility. In particular, equity-based pay with vesting periods, stock options in particular, could work as a retention tool for executives (e.g., Oyer, 2004; Jochem et al., 2018; Lie and Que, 2019). This mechanism implies that firms may want to use more equity-based pay when labor market mobility is high, to retain their executives.

Panel B of Figure 5 plots the average equity-based CEO pay (i.e., option and stock grants) as a fraction of total pay, along with our mobility measure. CEO compensation data cover

1936-2022 and are from [Frydman and Saks \(2010\)](#) and Execucomp.²¹ The panel shows that the increasing mobility pattern since the mid-1980s almost exactly coincides with the pattern of the average fraction of option grants over total CEO compensation. This positive correlation is consistent with option-based incentives becoming more important during the 1980s and 1990s in response to increasing executive mobility. We also note that the tax treatment of options pay changed for firms around 2003, as noted by [Murphy \(2013\)](#). As such, options pay became quantitatively less important than stock pay. We interpret the fall in options pay as largely coming from this tax change as opposed to a driver of mobility. Equity pay overall, though it peaked around 2000, has remained high while mobility has fallen.

In Panels C and D, we study the relation between pay, firm size and mobility. Panel C shows the pay-size ratio (total pay to firm size) and Panel D shows pay-size elasticity (how much pay increases for a 1% increase in size, as measured in 5-year rolling windows). These measures appeal to the competitive assignment literature ([Gabaix and Landier, 2008](#)). As pointed out by [Frydman and Saks \(2010\)](#) and others, pay and size only became closely linked after the 1970s. We find a similar relation with respect to mobility. Both measures mirror the increase-then-decrease of mobility. This suggests that mobility based on size (and the co-linked increasing pay packages) was a major driver in the 1986-2000 era, but fell off in the 2001-2022 era, as both pay packages (as a proportion of size) and pay elasticities (with respect to size) similarly fell.

These findings provide an interesting perspective on papers such as [Terviö \(2008\)](#); [Gabaix and Landier \(2008\)](#) and the plethora of papers in the CEO-matching literature. They suggest that the connection between firm size and pay impact mobility, both when the pay-size relation is strongly positive and less so. However, these papers also incorporate the importance of general CEO skills; our findings suggest further that firm-specific skills are important too, so the relation may be more nuanced.

Lastly, in Panel E, we analyze the relation between mobility and pay dispersion. As noted by [Jochem et al. \(2018\)](#), executive pay dispersion in the US has fallen dramatically across time. This suggests, that the “outside option” channel of potentially-high paying jobs has also

²¹We thank Carola Frydman for making her dataset available on her website. Pay data from Execucomp are adjusted for backfilling following [Gillan et al. \(2020\)](#).

become less important, which may have an impact on executive mobility. We find that this appears to be the case, when pay dispersion is high, mobility is high, and vice-versa.

In Table 7, we engage in time series analysis, and find that the visual patterns are mostly supported, even after controlling for important macro controls and time trends. In particular, equity share was a major driver in the 1986-2000 era, but has fallen off in the later era. The pay-size ratio, pay-size elasticity and pay dispersion measures both track executive mobility closely across both of the later eras.

4.6. Robustness: The Impact of Executives Moving to Private Firms

A potential explanation for the measured trends in executive mobility concerns a changing fraction of executives moved from public to private firms, which we do not observe because private firms are not accounted for in our data. For example, the decline in the executive mobility measures during the 2000s may be due to an increasing fraction of executives moving from public to private firms. To investigate this possibility, we randomly select approximately 10% of the final observed year of firm-CEO pairs in our sample and track out-of-our-sample career paths of these CEOs after their departure using Marquis Who's Who (which contains information on private firm employment).

We first confirm that this alternative data source also indicates increasing propensities for public firm CEOs to move to public firms over the past century (as we find in our main sample). During the 1920-1949, 1950-1985, and 1986-2011 periods, 1.6%, 4.2%, and 6.3% of departing CEOs moved to public companies as executives. These trends align with our main findings. Important for this robustness check, Who's Who allows us to document the rate at which departing public firm CEOs move to private firms, which has increased from 3.3%, 4.5%, and to 5.5% across the three eras.

Importantly, we do not find evidence that the recent decline in mobility is due to an increase in the public to private firm moves of CEOs. We find that the rate of public firm CEOs moving to public firms is 8.2% (3.4%) during the 1986-1999 (2000-2011) period, confirming the pattern in our data. Also, the rate of public firm CEOs moving to private firms only slightly increases (from 5.3% to 5.8%) across the periods. Therefore, an increasing propensity of move-

ments from public to private firms does not account for the decline in our mobility measures after 2000.²²

5. Cross-Sectional Determinants of Executive Mobility

The previous two sections document time-series trends in aggregate executive mobility over the past century and propose plausible explanations for the trends. In this section, we explore drivers of executive mobility in the cross-section of firms. This analysis sheds light on whether drivers of time-series mobility also explain across-firm heterogeneity, including which firm and CEO characteristics (such as firm size and CEO tenure) are associated with labor market mobility. To address these questions, we estimate the following logistic regression using a panel of firms from 1921-2022:²³

$$\mathbb{1} [Move]_{ijt} = \alpha_t + \gamma \cdot X_{ijt} + \varepsilon_{ijt} \quad (4)$$

where $\mathbb{1} [Move]_{ijt}$ is an indicator that is equal to one when the CEO of firm i in industry j as of year t moved to another firm in our sample by year $t+1$ or $t+2$ (i.e., within next two years), and zero otherwise; α_t represents year fixed effects; X_{ijt} represents a vector of firm, CEO, and industry characteristics including firm ROA, Tobin's Q , size (measured by log total assets), CEO tenure, and the two-digit SIC industry average and standard deviation of ROA for firm i in industry j in year t ; and ε_{ijt} represents random errors double clustered both at the two-digit SIC industry and year levels. All variables are Winsorized at the 1% and 99% levels to mitigate potential influences of outliers.

Table 8, presents estimation results for Eq. (4) based on the full sample (columns 1-3), and for a subsample of firm-years in which there is CEO departure (columns 4-6). Thus, columns 1-3 estimate the cross-sectional determinants of CEO mobility for sitting CEOs, similar to the mobility measure (A) in Eq. (2), which scales the number of CEO moves by the lagged number of CEOs. Columns 4-6 shows the cross-sectional determinants of CEO mobility conditional

²²It is also possible that if CEOs are on average older in the 2000-2011 period (because for example they continue to work later in life), then mobility might decline. However, we do not find evidence of this possibility: average ages of CEOs differ only slightly between 1986-1999 (55.9) and 2000-2011 (58.5).

²³We obtain quantitatively similar results using a linear probability model.

on turnover, similar to the first component of the mobility measure (B) in Eq. (2), representing the propensity of departing executives to find new executive work.

These regressions document several cross-sectional patterns of CEO mobility. First, consistent with CEO reallocation being more frequent when the benefits are greater, as shown above for aggregate mobility, column 2 in the two panels shows that the coefficients on the across-firm standard deviation of ROA in a given industry are significantly positive. Second, firm performance measured by ROA and Tobin's Q is negatively associated with the CEO's propensity to move to other firms as executive. In addition, industry-level average ROA is also negatively correlated with CEO mobility. These results are consistent with our time-series finding that executives tend to move in "bad times" (Table 5). Third, the coefficients on firm size are significantly positive in Panel A, suggesting that sitting CEOs of larger companies exhibit higher mobility. This result supports the assumption that larger firms offer better career opportunities. Lastly, the negative coefficients on CEO tenure suggest that short-tenure CEOs are more likely to move across firms than long-tenure CEOs.

Columns 7-9 presents estimation results for a version of Eq. (4) that uses an indicator for CEO turnover as dependent variable on the full sample. Thus, the panel shows the cross-sectional determinants of CEO turnover, similar to the second component of the mobility measure (C) in Eq. (2). The estimates confirm that the portion of CEO mobility due to turnover largely reinforces the cross-sectional patterns we find in Panels A and B — CEOs depart more when the benefits of reallocation are larger and the firm or industry performs worse; and larger firm CEOs experience more churning in the labor market. Consistent with the cross-sectional mobility results for CEOs capturing patterns of executive mobility in general, we find quantitatively similar results when we estimate Eq. (4) using CFO moves and turnovers (untabulated). Overall, the evidence in Table 8 shows that firm and industry performance and its dispersion are key determinants of mobility in the cross-section, consistent with our time-series evidence.

6. Conclusion

We uncover several new long-run trends in the job mobility of executives by constructing a new dataset of executive movements during 1920-2022. CEO and CFO mobility generally increased over the six decades leading up to the end of the 20th century. This increase was accompanied by executives moving across an increasingly diverse set of industries over time. In stark contrast, following an inflection point around 2000, executive mobility declined sharply, returning to levels not seen since the 1980s.

We offer several explanations for the evolution of mobility over the century, including changing benefits to reallocating executives across firms, labor market size, and importance of general managerial ability. A key takeaway is that the benefits to executive reallocation are significantly positively associated with aggregate mobility both when mobility was increasing (before 2000) and also when mobility was decreasing (after 2000). Future research should explore the ramifications of the decreasing executive mobility for corporate policies, as well as why executive movements across firms appear more efficient relative to reallocation of rank-and-file employees and capital assets.

We also document substantial time-series co-movement between executive mobility and equity-based executive pay, and investigate whether the relation is due to firms' attempt to retain executives when their mobility increases.

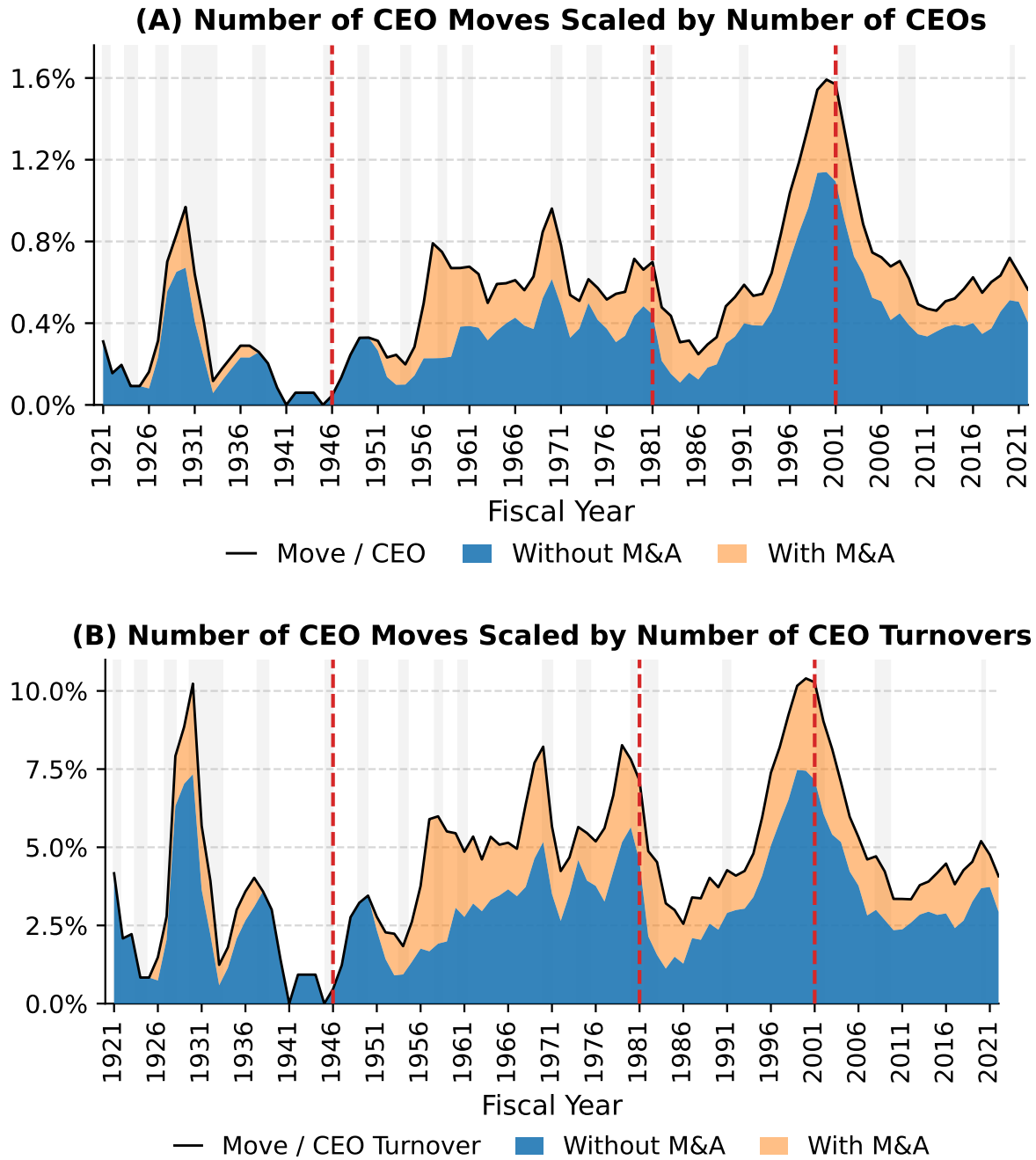
References

- Barry, J. W., Lyman, N., and Zhao, L. (2025). Human capital, competition and mobility in the managerial labor market. Working Paper.
- Covarrubias, M., Gutiérrez, G., and Philippon, T. (2019). *From Good to Bad Concentration? U.S. Industries Over the Past 30 Years*, chapter 4, pages 1–46. University of Chicago Press.
- Custódio, C., Ferreira, M. A., and Matos, P. (2013). Generalists versus specialists: Lifetime work experience and chief executive officer pay. *Journal of Financial Economics*, 108(2):471–492.
- Cziraki, P. and Jenter, D. (2022). The market for ceos. *Working Paper*.
- Decker, R. A., Haltiwanger, J., Jarmin, R. S., and Miranda, J. (2020). Changing business dynamism and productivity: Shocks versus responsiveness. *American Economic Review*, 110(12):3952–3990.
- Diamond, P. A. (1982). Aggregate demand management in search equilibrium. *Journal of Political Economy*, 90(5):881–894.
- Doidge, C., Karolyi, G. A., and Stulz, R. M. (2017). The u.s. listing gap. *Journal of Financial Economics*, 123(3):464–487.
- Eisfeldt, A. and Rampini, A. (2006). Capital reallocation and liquidity. *Journal of Monetary Economics*, 53(3):369–399.
- Frydman, C. (2019). Rising through the ranks: The evolution of the market for corporate executives, 1936–2003. *Management Science*, 65(11):4951–4979.
- Frydman, C. and Saks, R. E. (2010). Executive compensation: A new view from a long-term perspective, 1936–2005. *Review of Financial Studies*, 23(5):2099–2138.
- Gabaix, X. and Landier, A. (2008). Why has ceo pay increased so much? *The quarterly journal of economics*, 123(1):49–100.
- Gartner (2019). Top 10 emerging skills for the c-suite. <https://www.gartner.com/smarterwithgartner/top-10-emerging-skills-for-the-c-suite>. Accessed: 2025-07-02.
- Gillan, S., Hartzell, J. C., Koch, A., and Starks, L. T. (2020). Getting the incentives right: Backfilling and biases in executive compensation data. *Review of Financial Studies*, 31(4):1460–1498.
- Graham, J. R., Kim, H., and Leary, M. (2020). Ceo-board dynamics. *Journal of Financial Economics*, 137(3):612–636.
- Groysberg, B., Kelly, L. K., and MacDonald, B. (2011). The new path to the c-suite. *Harvard Business Review*.
- Grullon, G., Larkin, Y., and Michaely, R. (2019). Are us industries becoming more concentrated? *Review of Finance*, 23(4):697–743.
- Harmon, N. A. (2013). Are workers better matched in large labor markets? Working paper, University of Copenhagen.

- Huson, M. R., Parrino, R., and Starks, L. T. (2001). Internal monitoring mechanisms and ceo turnover: A long-term perspective. *The Journal of Finance*, 56(6):2265–2297.
- Jochem, T., Ladika, T., and Sautner, Z. (2018). The retention effects of unvested equity: Evidence from accelerated option vesting. *Review of Financial Studies*, 31(11):4142–4186.
- Jochem, T., Ormazabal, G., and Rajamani, A. (2025). Why have ceo pay levels become less diverse? *Journal of Finance*. Forthcoming.
- Kankanhalli, G., Kim, H., and Kim, H. (2025). Asset redeployability across time. Working paper.
- Kaplan, S. N. and Minton, B. A. (2012). How has ceo turnover changed? *International Review of Finance*, 12(1):57–87.
- Kim, H. and Kung, H. (2017). The asset redeployability channel: How uncertainty affects corporate investment. *The Review of Financial Studies*, 30(1):245–280.
- Kini, O., Williams, R., and Yin, S. (2020). Ceo noncompete agreements, job risk, and compensation. *The Review of Financial Studies*, 34(10):4701–4744.
- Lie, E. and Que, T. (2019). On the use of option grants as a retention tool. *Available at SSRN 3504794*.
- Moretti, E. and Yi, M. (2024). Size matters: Matching externalities and the advantages of large labor markets. Technical report, National Bureau of Economic Research.
- Murphy, K. J. (1999). Executive compensation. In Ashenfelter, O. C. and Card, D., editors, *Handbook of Labor Economics*. Elsevier, North Holland.
- Murphy, K. J. (2013). Executive compensation: Where we are, and how we got there. In Constantinides, G. M., Harris, M., and Stulz, R. M., editors, *Handbook of the Economics of Finance*, volume 2, Part A, pages 211–356. Elsevier Science, North Holland.
- Murphy, K. J. and Zabojnik, J. (2007). Managerial capital and the market for ceos. *Available at SSRN 984376*.
- Oyer, P. (2004). Why do firms use incentives that have no incentive effects? *Journal of Finance*, 59(4):1619–1649.
- Petrongolo, B. and Pissarides, C. (2006). Scale effects in markets with search. *The Economic Journal*, 116(508):21–44.
- Saks, R. E. and Wozniak, A. (2011). Labor reallocation over the business cycle: New evidence from internal migration. *Journal of Labor Economics*, 29(4):697–739.
- Terviö, M. (2008). The difference that ceos make: An assignment model approach. *American Economic Review*, 98(3):642–668.
- Vancil, R. F. (1987). *Passing the Baton: Managing the Process of CEO Succession*. Harvard Business School Press, Boston, MA.

Figure 1. Measures of Executive Mobility for CEOs

This figure plots three measures of executive mobility for CEOs from 1921 through 2021 (three-year moving average). Move is the number of CEOs in year $t - 2$ or $t - 1$ who become CEOs or other executive officers at other firms in our sample in year t . CEO is the total number CEOs in year $t - 1$. Turnover is the number of CEOs who leave firms in year $t - 1$. All Moves includes all CEO movements while Moves without M&As excludes M&A or subsidiary-related moves. Moves with M&As is M&A or subsidiary-related moves. Panel C decomposes (A) Move/CEO into (B) Move/Turnover and (C) Turnover/CEO as in Eq. (2) and shows how each component evolves over time.



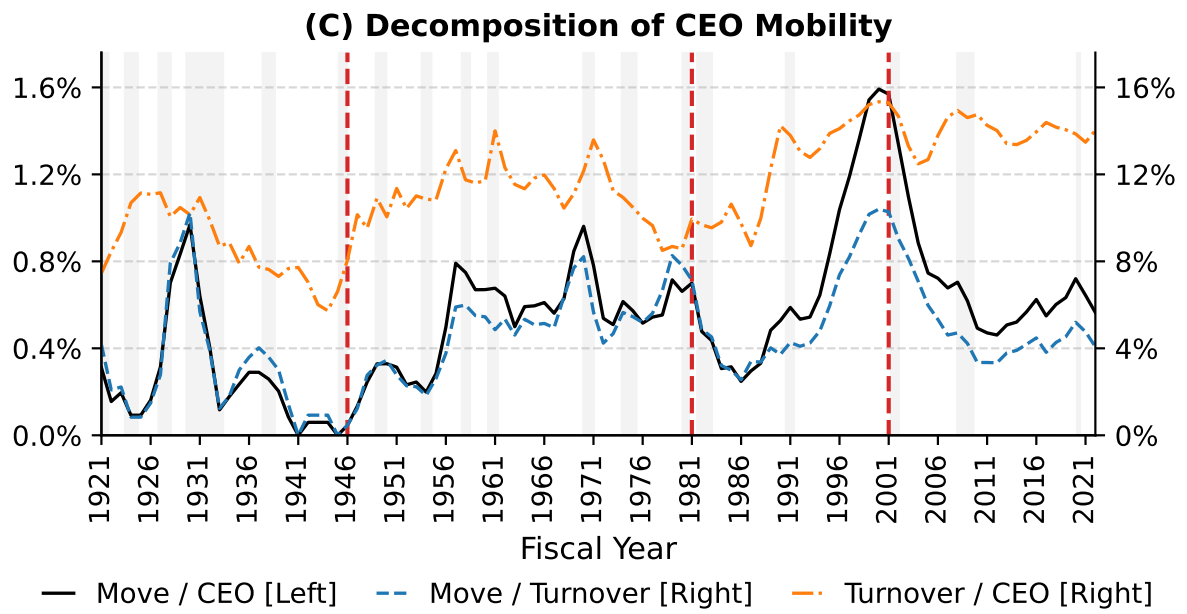
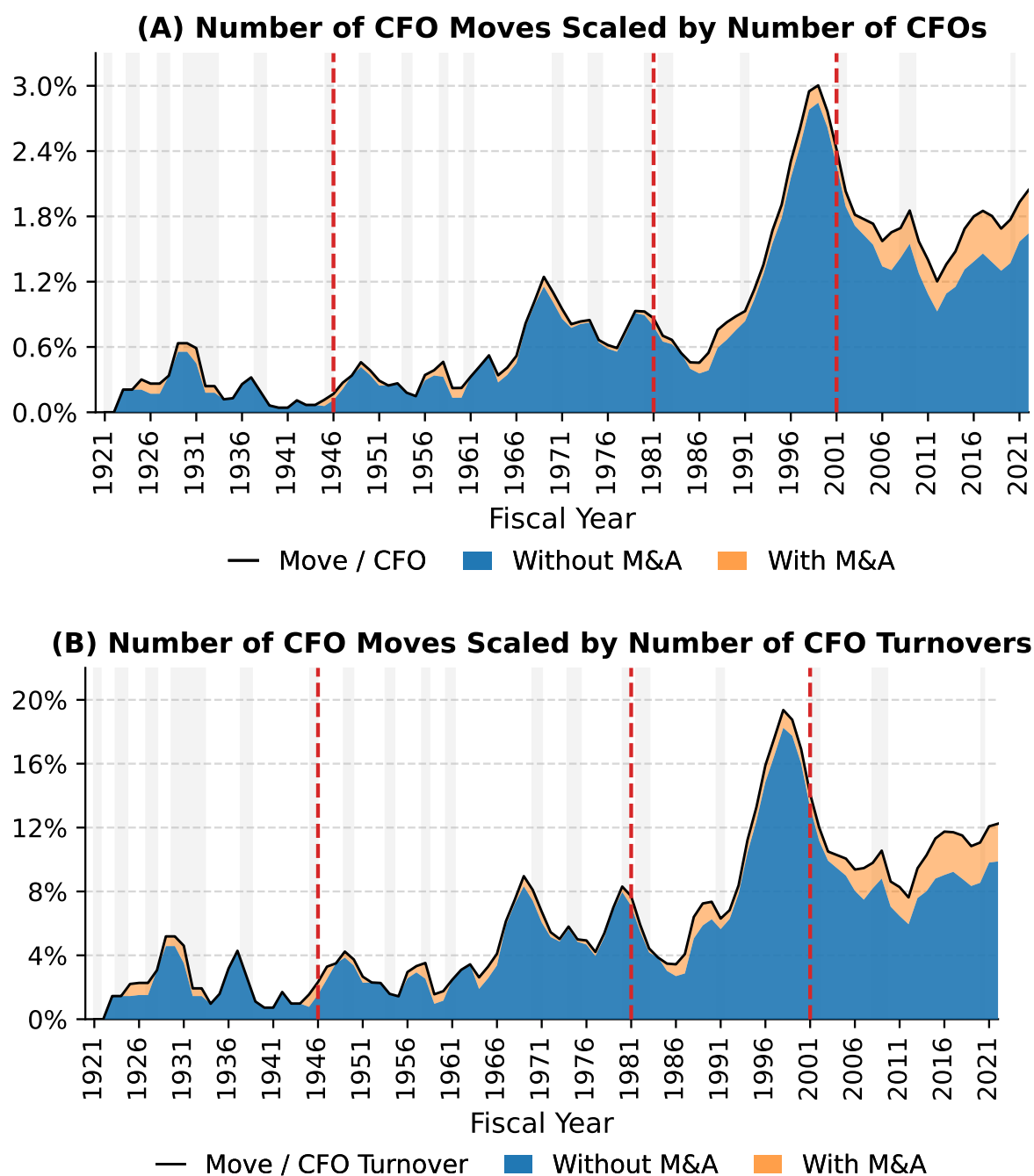


Figure 2. Measures of Executive Mobility for CFOs

This figure plots three measures of executive mobility for CFOs from 1921 through 2021 (three-year moving average). Move is the number of CFOs in year $t-2$ or $t-1$ who become CFOs or other executive officers at other firms in our sample in year t . CFO is the total number CFOs in year $t-1$. Turnover is the number of CFOs who leave firms in year $t-1$. All Moves includes all CFO movements while Moves without M&As excludes M&A or subsidiary-related moves. Moves with M&As is M&A or subsidiary-related moves. Panel C decomposes (A) Move/CFO into (B) Move/Turnover and (C) Turnover/CFO as in Eq. (2) and shows how each component evolves over time.



(C) Decomposition of CFO Mobility

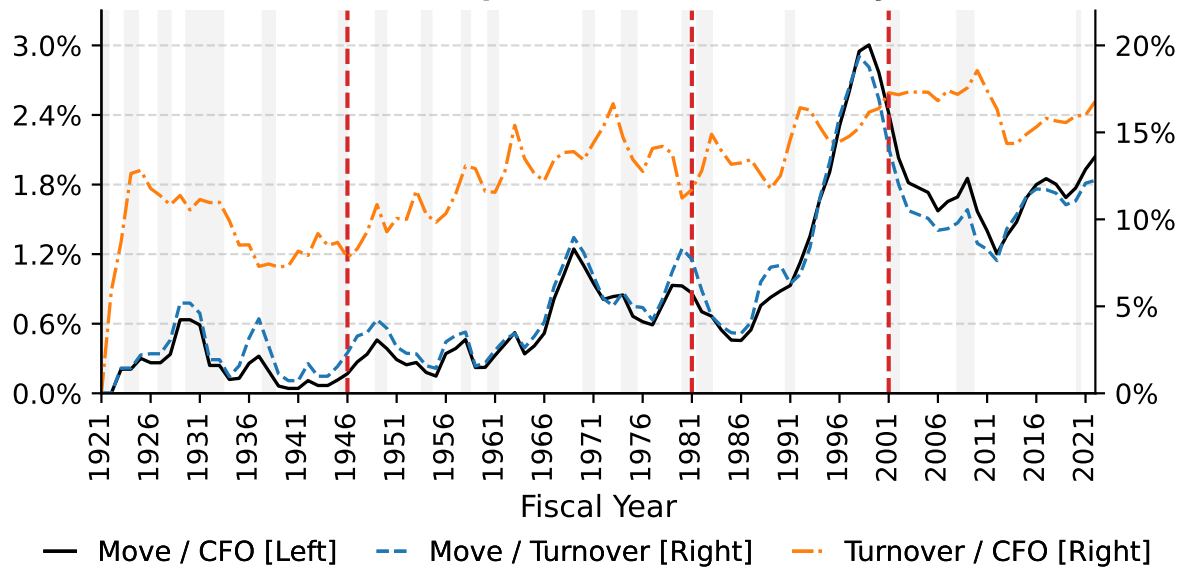
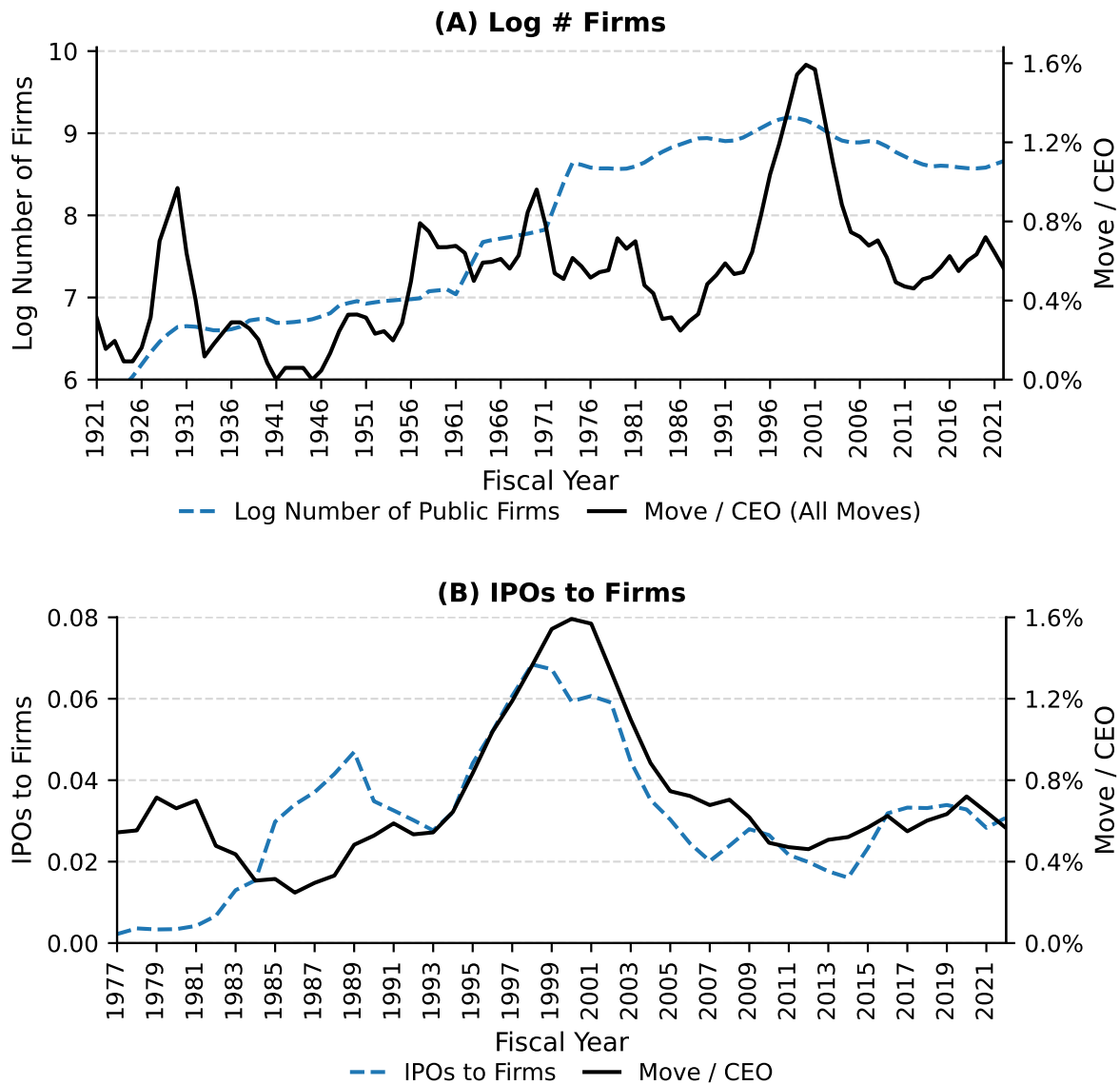


Figure 3. Executive Mobility and Underlying Factors

This figure compares the trends in variables that potentially explain executive mobility with our mobility measure (three-year moving average). Move is the number of CEOs in year $t - 2$ or $t - 1$ who become CEOs or other executive officers at other firms in our sample in year t . CEO is the total number CEOs in year $t - 1$. All Moves includes all CEO movements while Moves without M&As excludes M&A or subsidiary-related moves. Panel A shows the log number of NYSE, Amex and Nasdaq listed public firms. Panel B shows the standard deviation of NYSE and Amex listed firms' return on assets. Panel C shows the inverse of the Herfindahl-Hirschman Index of the fraction of CEO moves across one-digit SIC industries over the past ten years. Panel D shows the number of business master's degree programs enrollment over the number of engineering master's degree programs enrollment in the US. Panel E shows the number of completed M&A deals by US public acquirors over the number of NYSE, Amex and Nasdaq listed public firms. Panel F shows the fraction of equity pay (stock and option grants) and option grants over total pay (salary and bonus plus equity pay). Panel G shows the fraction of across-state and across-county CEO moves over the past ten years.



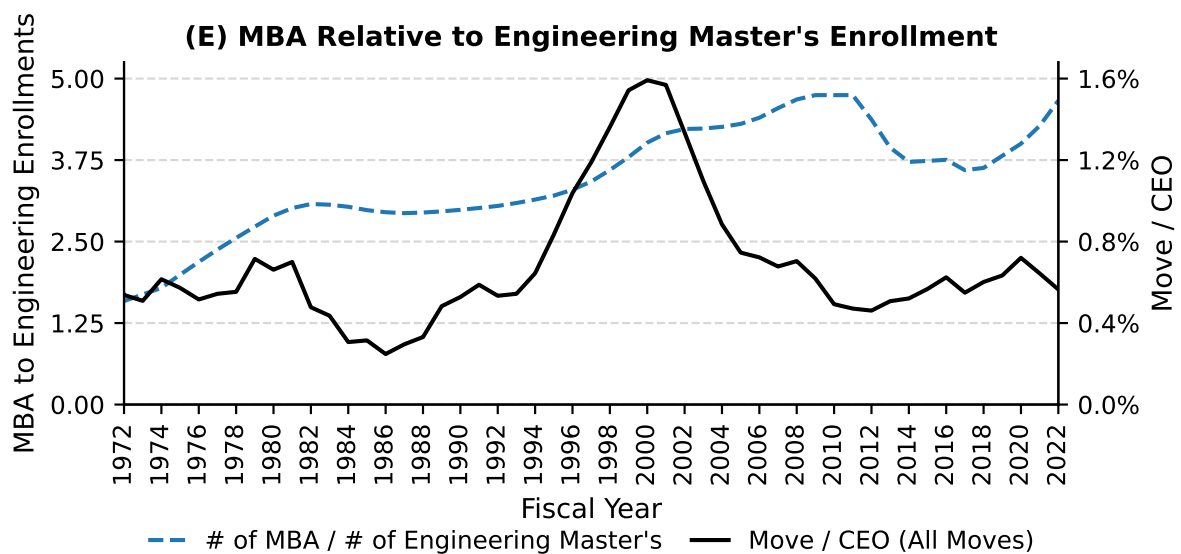
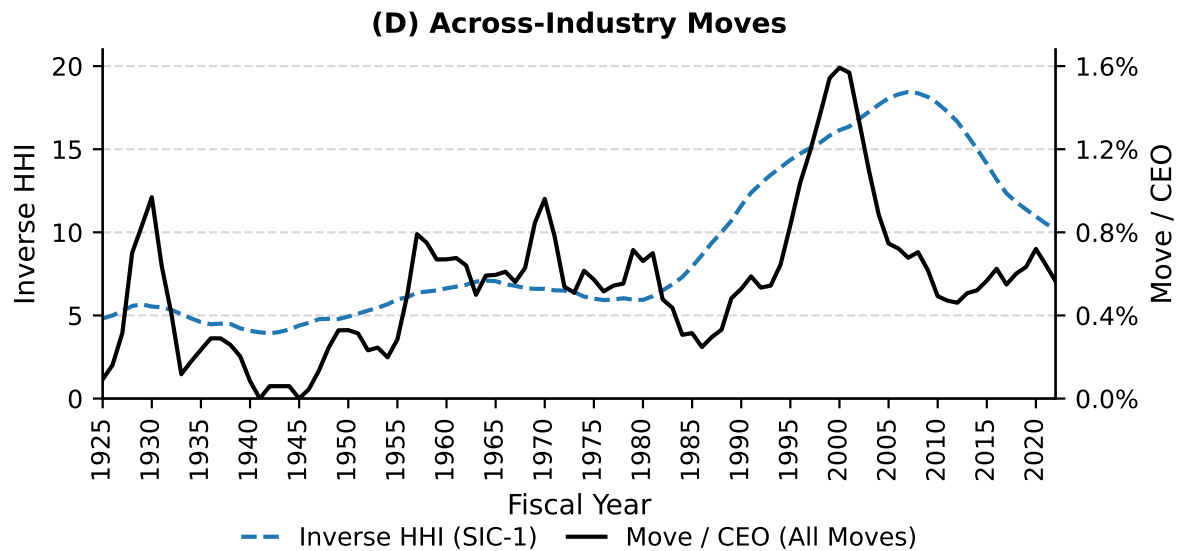
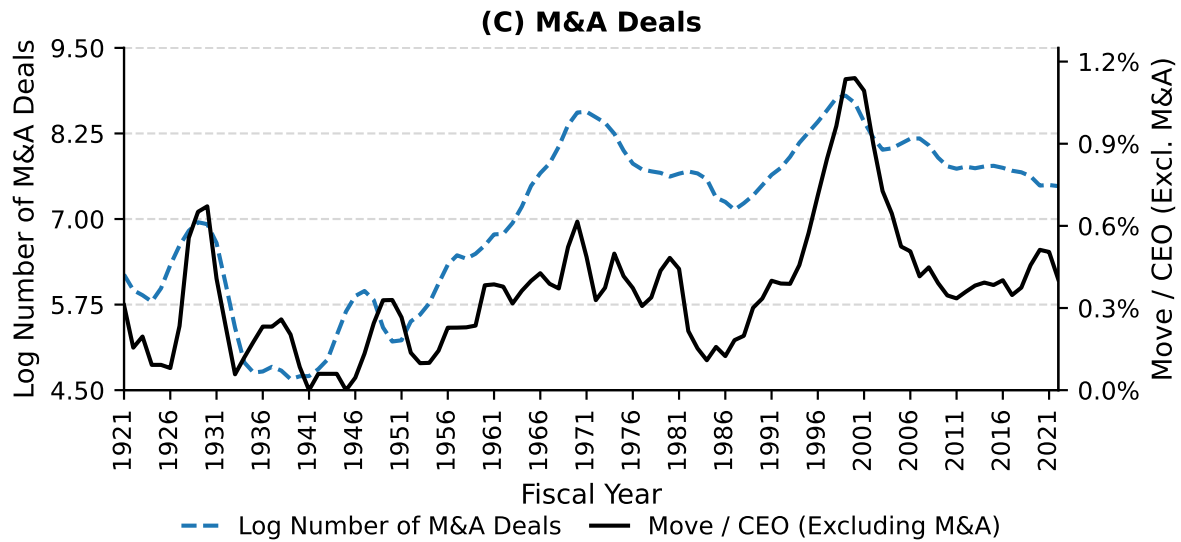


Figure 4. Executive Mobility and Mobility of the Factors of Production

This figure displays the relationship between executive (CEO) mobility and both asset and labor redeployability. Asset redeployability comes from [Kankanhalli et al. \(2025\)](#). Labor redeployability is measured as the inverse-HHI of census moves across IND1990 industries in Census microdata. When labor redeployability is high, workers in the economy move to a broader range of industries. Asset redeployability is available from 1921, labor redeployability is available from 1976 (the year at which census individual IDs became available and individual-level moves across industry became measurable).

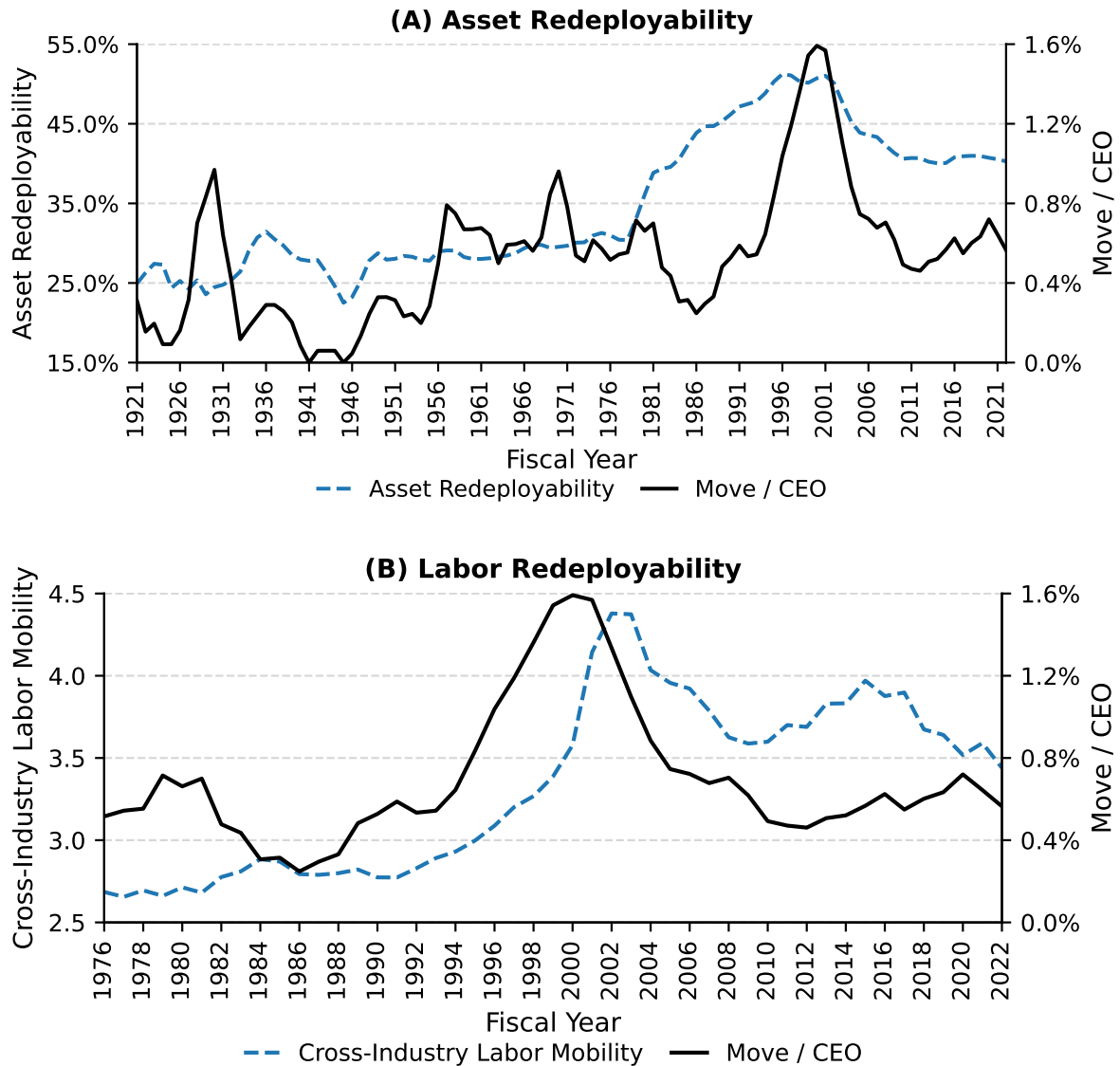
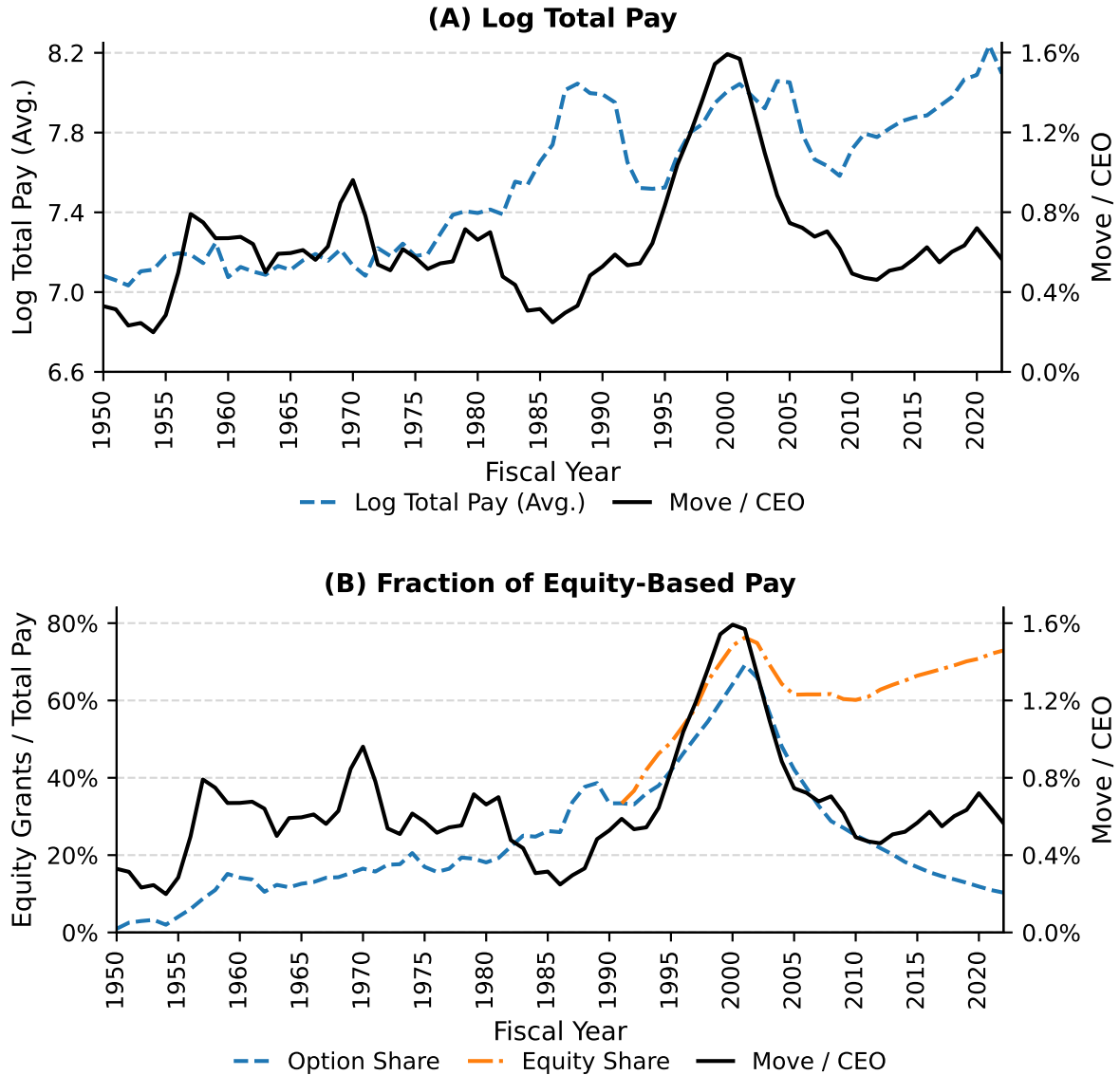


Figure 5. Executive Mobility and Compensation

This figure compares trends in executive mobility and trends in executive compensation. Panel A displays average log total pay. Panel B displays both the fraction of option-based pay (Frydman and Saks, 2010) along with the fraction of equity-based pay. Panel C displays the pay to market capitalization ratio. Panel D displays pay-size elasticity, which we measure as regressions of log pay on log size in rolling 5 year windows. Panel E displays pay dispersion, which is the difference between the 90th and 50th percentile of pay in a given year (Jochem et al., 2025).



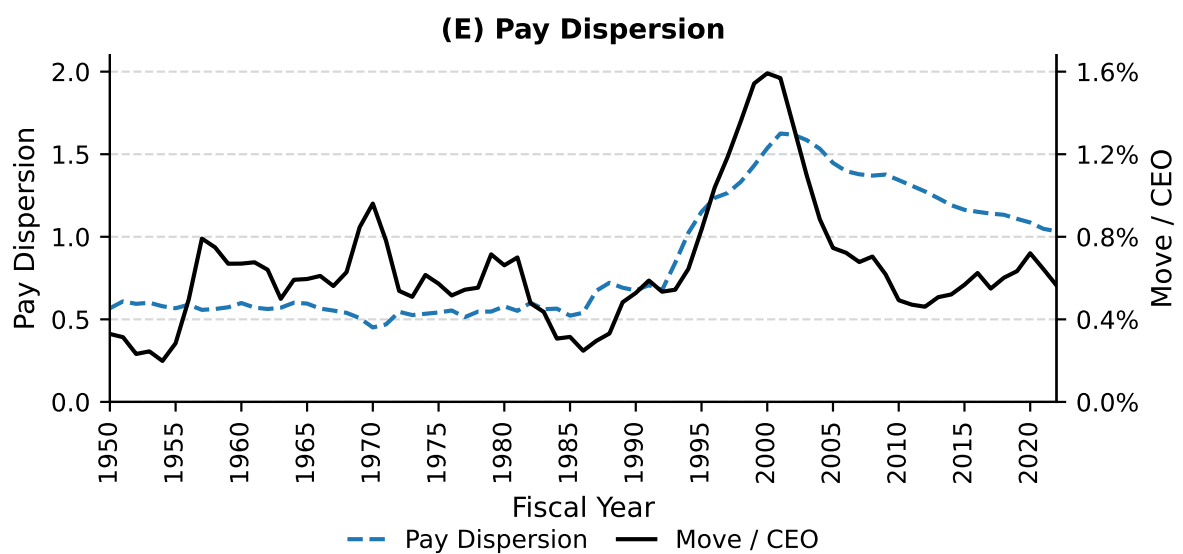
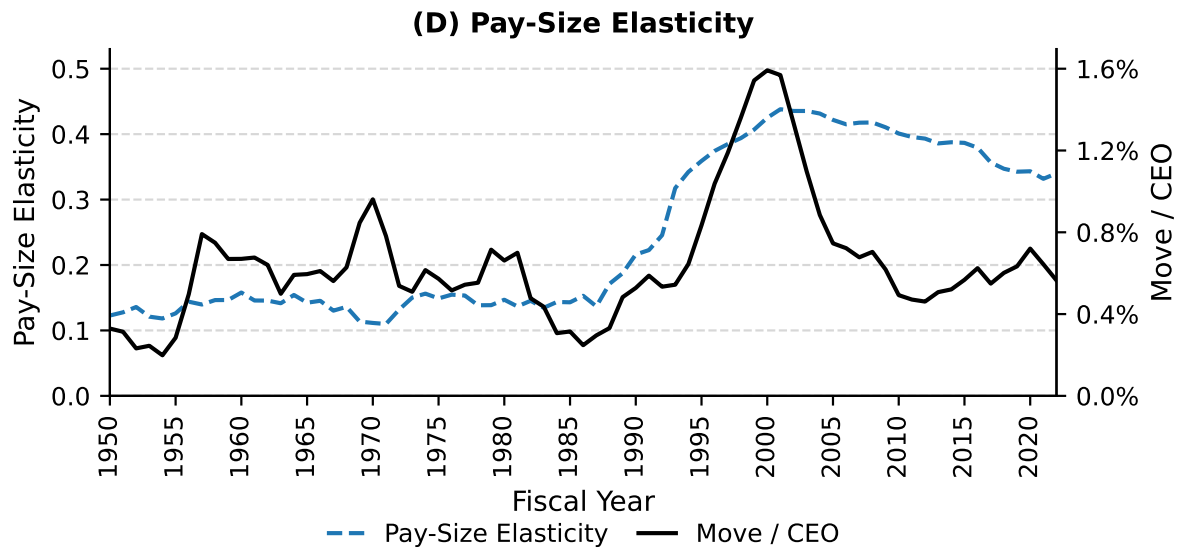
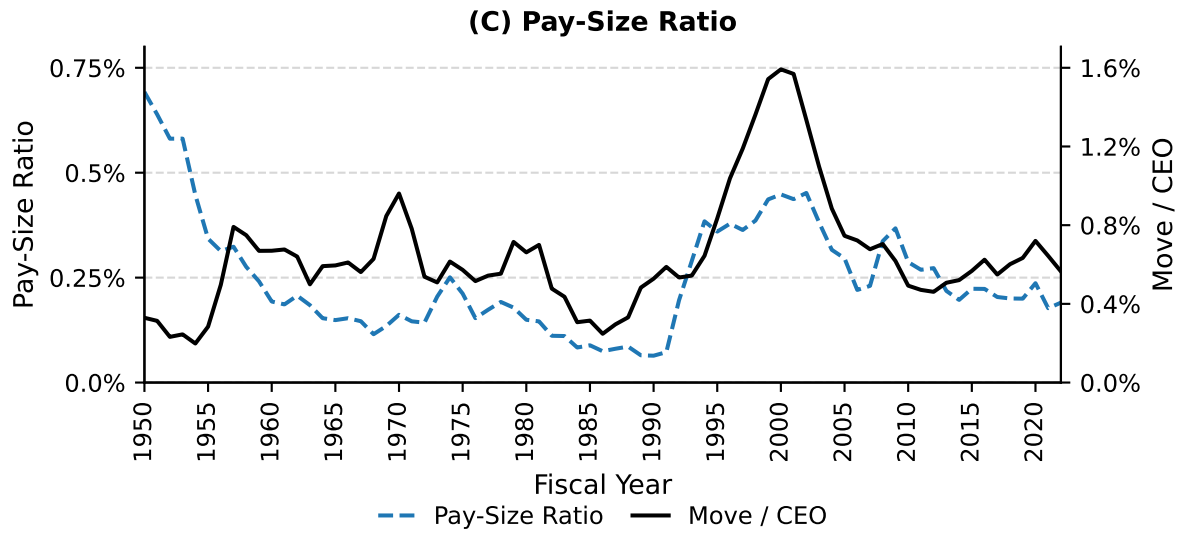


Table 1. CEO and CFO Moves and New Job Titles

This table presents the number of CEOs and CFOs who leave office (Turnovers) and the number of CEOs and CFOs who are hired at new firms (Moves) in our sample from 1920 through 2011 in three different periods. A move is defined as a CEO (CFO) who worked at a firm in year $t - 2$ or $t - 1$ moves to another firm in year t . The new title is the first new job title of a former CEO (CFO) who moves to a new firm in Panel A (Panel B), with Non-CEO (Non-CFO) indicating a move to a non-CEO (non-CFO) role at the new firm (e.g., COO). Numbers in parentheses are the ratio of the number of former CEOs or CFOs who move to other firms to the number of CEO or CFO turnovers in each period.

Panel A. CEO Turnovers, Moves and New Job Titles

	(1)	(2)	(3)	(4)	(5)
	Full Sample	1920-1945	1946-1985	1986-2000	2001-2022
CEO Turnovers	35,366	1,113	5,270	9,969	19,014
Officer of new firm (%)	1,900 (5.37%)	35 (3.14%)	271 (5.14%)	719 (7.21%)	875 (4.60%)
CEO of new firm (%)	837 (2.37%)	14 (1.26%)	62 (1.18%)	276 (2.77%)	485 (2.55%)
Non-CEO officer of new firm (%)	1,063 (3.01%)	21 (1.89%)	209 (3.97%)	443 (4.44%)	390 (2.05%)

Panel B. CFO Turnovers, Moves and New Job Titles

	(1)	(2)	(3)	(4)	(5)
	Full Sample	1920-1945	1946-1985	1986-2000	2001-2022
CFO Turnovers	34,397	1,105	6,161	9,242	17,889
Officer of new firm (%)	3,503 (10.18%)	30 (2.71%)	300 (4.87%)	1,270 (13.74%)	1,903 (10.64%)
CFO of new firm (%)	2,813 (8.18%)	15 (1.36%)	167 (2.71%)	977 (10.57%)	1,654 (9.25%)
Non-CFO officer of new firm (%)	690 (2.01%)	15 (1.36%)	133 (2.16%)	293 (3.17%)	249 (1.39%)
CEO of new firm (%)	175 (0.51%)	0 (0.00%)	10 (0.16%)	80 (0.87%)	85 (0.48%)

Table 2. Relative Firm Size and Compensation for Moving CEOs and CFOs

This table presents the relative firm size and compensation change for CEOs and CFOs who move to other firms from 1920 through 2011. A move is defined as a CEO (CFO) who worked at a firm in year $t - 2$ or $t - 1$ moves to another firm in year t . New Title is the first job title of an externally hired former CEO (CFO). To larger firms is the fraction of former CEOs or CFOs who move to a firm that is larger than their previous firm in terms of total assets. Pay change is the percentage change in an externally hired CEO/CFO salary and bonus plus option and stock grants in the first year, relative to that from her previous employment. We do not present pay change results before 1986 given the small number of observations for which we can observe pay both before and after moves. Changes in pay are winsorized at the 5th and 95th percentiles. The numbers in parentheses are the number of observations underlying the percentages.

	(1)	(2)	(3)	(4)	(5)
	Full Sample	1920-1945	1946-1985	1986-2000	2001-2022
To larger firm					
CEO moves					
All	62.20%	68.57%	71.84%	65.05%	57.55%
	(1,651)	(35)	(174)	(641)	(801)
CEO	51.23%	71.43%	51.22%	51.23%	50.51%
	(691)	(14)	(41)	(244)	(392)
Non-CEO	70.10%	66.67%	78.20%	73.55%	64.30%
	(960)	(21)	(133)	(397)	(409)
CFO moves					
All	50.82%	65.52%	49.79%	47.37%	53.04%
	(3,156)	(29)	(237)	(1,161)	(1,729)
CFO	50.99%	60.00%	49.25%	46.52%	53.74%
	(2,524)	(15)	(134)	(892)	(1,483)
Non-CFO	50.16%	71.43%	50.49%	50.19%	48.78%
	(632)	(14)	(103)	(269)	(246)
Pay change					
CEO moves					
All	125.18%			180.74%	111.18%
	(394)			(152)	(238)
CEO	76.15%			91.23%	72.44%
	(192)			(62)	(130)
Non-CEO	198.13%			237.80%	155.21%
	(202)			(90)	(108)
CFO moves					
All	98.87%			218.69%	58.95%
	(764)			(228)	(531)
CFO	94.60%			215.01%	53.47%
	(630)			(174)	(454)
Non-CFO	141.61%			216.17%	83.34%
	(134)			(54)	(77)

Table 3. Frequency of CEO Moves between Industries

This table presents the frequency of CEOs moving from a firm in the origination industry to another firm in the destination industry for three time periods from 1920 through 2011. A move is defined as a CEO who worked at a firm in year $t - 2$ or $t - 1$ moves to another firm in year t . SIC-From is the one-digit SIC industry of the firm that a given CEO departs and SIC-To is the one-digit SIC industry of the firm to which the CEO moves as CEO or another executive. HHI is the Herfindahl-Hirschman Index of the fraction of moves across one-digit SIC industries. The mean fractions are calculated using cells with non-zero values, while the HHI is calculated by assuming zeros for cells with missing values. Across-industry move is the fraction of out-of-industry moves among all moves in a given period.

Indicates top 10% top 25% top 50% industry pairs in terms of frequency of moves within era.

Panel A. 1920-1945: 35 moves (Mean = 7.69%, HHI = 17.55%, Across-industry move = 37.14%)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	SIC0	SIC1	SIC2	SIC3	SIC4	SIC5	SIC6	SIC7	SIC8	SIC9	Total
SIC0. Agriculture, Forestry, Fishing											
SIC1. Mining & Construction						2.86%					2.86%
SIC2. Light Manufacturing		2.86%	22.86%		2.86%	2.86%					31.43%
SIC3. Heavy Manufacturing		2.86%	8.57%	31.43%			2.86%	2.86%			48.57%
SIC4. Transportation & Public Utilities											
SIC5. Wholesale & Retail Trade			2.86%			8.57%					11.43%
SIC6. Finance, Insurance, Real Estate		5.71%									5.71%
SIC7. General Services											
SIC8. Professional Services											
SIC9. Public Administration											
Total		11.43%	34.29%	31.43%	2.86%	14.29%	2.86%	2.86%			100.00%

Panel B. 1946-1985: 271 moves (Mean = 2.50%, HHI = 15.75%, Across-industry move = 46.86%)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	SIC0	SIC1	SIC2	SIC3	SIC4	SIC5	SIC6	SIC7	SIC8	SIC9	Total
SIC0. Agriculture, Forestry, Fishing											
SIC1. Mining & Construction		1.11%	0.74%	1.85%		0.74%			0.37%		4.80%
SIC2. Light Manufacturing		2.21%	11.44%	8.49%	0.37%	1.48%	0.37%	1.11%			25.46%
SIC3. Heavy Manufacturing		0.74%	8.12%	35.06%	0.74%	4.43%	1.48%	0.37%	0.74%		51.66%
SIC4. Transportation & Public Utilities		0.37%			0.37%	0.74%		0.37%			1.85%
SIC5. Wholesale & Retail Trade		0.37%	2.58%	3.69%		3.69%		0.37%	0.37%		11.07%
SIC6. Finance, Insurance, Real Estate			0.37%		0.37%	0.37%			0.37%		1.48%
SIC7. General Services			0.74%	0.37%		0.37%		1.48%			2.95%
SIC8. Professional Services			0.37%					0.37%			0.74%
SIC9. Public Administration											
Total		4.80%	24.35%	49.45%	1.85%	11.81%	1.85%	4.06%	1.85%		100.00%

Panel C. 1986-2000: 719 moves (Mean = 1.69%, HHI = 6.64%, Across-industry move = 47.98%)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	SIC0	SIC1	SIC2	SIC3	SIC4	SIC5	SIC6	SIC7	SIC8	SIC9	Total
SIC0. Agriculture, Forestry, Fishing				0.14%				0.28%			0.42%
SIC1. Mining & Construction	0.14%	4.45%	0.70%	0.83%	0.28%	0.14%	0.14%	0.42%	0.14%		7.23%
SIC2. Light Manufacturing	0.28%	0.28%	5.70%	3.76%	1.11%	0.97%		0.70%	1.11%		13.91%
SIC3. Heavy Manufacturing	0.14%	0.70%	2.64%	19.19%	2.23%	2.36%		3.34%	1.39%		31.99%
SIC4. Transportation & Public Utilities		0.14%		1.11%	3.48%	0.14%		1.25%			6.12%
SIC5. Wholesale & Retail Trade			0.56%	2.64%	0.14%	6.68%	0.14%	1.81%	0.97%		12.93%
SIC6. Finance, Insurance, Real Estate		0.28%						0.42%	0.28%		0.97%
SIC7. General Services		0.14%	0.42%	2.50%	1.53%	1.53%		9.46%	1.25%	0.14%	16.97%
SIC8. Professional Services	0.14%	0.28%	1.95%	1.81%	0.14%	0.42%		1.53%	3.06%		9.32%
SIC9. Public Administration									0.14%		0.14%
Total	0.70%	6.26%	11.96%	31.99%	8.90%	12.24%	0.28%	19.19%	8.34%	0.14%	100.00%

Panel D. 2001-2022: 871 moves (Mean = 1.59%, HHI = 6.18%, Across-industry move = 42.94%)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	SIC0	SIC1	SIC2	SIC3	SIC4	SIC5	SIC6	SIC7	SIC8	SIC9	Total
SIC0. Agriculture, Forestry, Fishing											
SIC1. Mining & Construction	0.11%	4.36%	0.34%	0.57%	0.57%	0.23%	0.34%		0.23%		6.77%
SIC2. Light Manufacturing	0.23%	0.57%	14.70%	2.64%	0.80%	1.38%	1.38%	0.92%	1.03%		23.65%
SIC3. Heavy Manufacturing		0.92%	2.64%	12.97%	0.80%	0.69%	0.80%	2.99%	0.69%		22.50%
SIC4. Transportation & Public Utilities		0.11%	0.80%	0.80%	4.48%	0.34%	0.34%	0.69%			7.58%
SIC5. Wholesale & Retail Trade		0.34%	0.57%	1.72%	0.57%	3.67%	0.80%	0.69%			8.38%
SIC6. Finance, Insurance, Real Estate	0.11%	0.69%	0.69%	0.92%	0.92%	0.34%	7.92%	0.69%	0.23%		12.51%
SIC7. General Services	0.11%	0.46%	1.03%	2.64%	0.69%	0.80%	1.15%	7.81%	0.46%		15.15%
SIC8. Professional Services			0.69%	0.69%	0.11%		0.46%	0.34%	1.15%		3.44%
SIC9. Public Administration											
Total	0.57%	7.46%	21.47%	22.96%	8.96%	7.46%	13.20%	14.12%	3.79%		100.00%

Table 4. Frequency of CFO Moves between Industries

This table presents the frequency of CFOs moving from a firm in the origination industry to another firm in the destination industry for three time periods from 1920 through 2011. A move is defined as a CFO who worked at a firm in year $t - 2$ or $t - 1$ moves to another firm in year t . SIC-From is the one-digit SIC industry of the firm that a given CFO departs and SIC-To is the one-digit SIC industry of the firm to which the CFO moves as CFO or another executive. HHI is the Herfindahl-Hirschman Index of the fraction of moves across one-digit SIC industries. The mean fractions are calculated using cells with non-zero values, while the HHI is calculated by assuming zeros for cells with missing values. Across-industry move is the fraction of out-of-industry moves among all moves in a given period.

Indicates top 10% top 25% top 50% industry pairs in terms of frequency of moves within era.

Panel A. 1920-1945: 30 moves (Mean = 7.69%, HHI = 14.44%, Across-industry move = 46.67%)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	SIC0	SIC1	SIC2	SIC3	SIC4	SIC5	SIC6	SIC7	SIC8	SIC9	Total
SIC0. Agriculture, Forestry, Fishing											
SIC1. Mining & Construction				6.67%			3.33%				10.00%
SIC2. Light Manufacturing		3.33%	23.33%	13.33%				3.33%			43.33%
SIC3. Heavy Manufacturing			3.33%	23.33%	3.33%						30.00%
SIC4. Transportation & Public Utilities											
SIC5. Wholesale & Retail Trade				3.33%		6.67%					10.00%
SIC6. Finance, Insurance, Real Estate		3.33%	3.33%								6.67%
SIC7. General Services											
SIC8. Professional Services											
SIC9. Public Administration											
Total		6.67%	30.00%	46.67%	3.33%	6.67%	3.33%	3.33%			100.00%

Panel B. 1946-1985: 300 moves (Mean = 2.33%, HHI = 10.95%, Across-industry move = 53.33%)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	SIC0	SIC1	SIC2	SIC3	SIC4	SIC5	SIC6	SIC7	SIC8	SIC9	Total
SIC0. Agriculture, Forestry, Fishing											
SIC1. Mining & Construction		1.00%	1.33%	2.00%		1.67%		0.33%			6.33%
SIC2. Light Manufacturing	0.33%	2.67%	13.00%	8.00%	0.67%	2.33%	0.67%	0.33%	0.33%		28.33%
SIC3. Heavy Manufacturing		3.00%	7.67%	26.33%	1.00%	3.33%	2.33%	1.33%	1.00%		46.00%
SIC4. Transportation & Public Utilities			0.33%	0.33%	0.33%	0.33%		0.33%			1.67%
SIC5. Wholesale & Retail Trade		0.33%	2.00%	4.00%		5.67%	0.33%	0.33%			12.67%
SIC6. Finance, Insurance, Real Estate		0.33%	0.33%	0.33%							1.00%
SIC7. General Services			0.33%	1.67%		0.33%	0.33%	0.33%			3.00%
SIC8. Professional Services				0.67%				0.33%			1.00%
SIC9. Public Administration											
Total	0.33%	7.33%	25.00%	43.33%	2.00%	13.67%	3.67%	3.33%	1.33%		100.00%

Panel C. 1986-2000: 1,270 moves (Mean = 1.64%, HHI = 6.24%, Across-industry move = 55.12%)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	SIC0	SIC1	SIC2	SIC3	SIC4	SIC5	SIC6	SIC7	SIC8	SIC9	Total
SIC0. Agriculture, Forestry, Fishing								0.08%			0.08%
SIC1. Mining & Construction		3.39%	0.55%	0.55%	0.55%	0.24%		0.55%	0.24%		6.06%
SIC2. Light Manufacturing	0.16%	0.16%	3.78%	3.39%	0.55%	1.10%	0.16%	2.13%	1.42%		12.83%
SIC3. Heavy Manufacturing	0.08%	0.79%	2.76%	17.01%	1.18%	2.13%	0.08%	8.35%	1.97%		34.33%
SIC4. Transportation & Public Utilities		0.08%	0.24%	1.02%	2.68%	0.39%		2.20%	0.24%	0.08%	6.93%
SIC5. Wholesale & Retail Trade		0.31%	1.10%	1.50%	0.55%	5.28%	0.08%	2.68%	0.55%		12.05%
SIC6. Finance, Insurance, Real Estate					0.08%	0.08%		0.16%			0.31%
SIC7. General Services		0.39%	0.94%	4.02%	0.71%	1.89%	0.16%	10.63%	1.57%		20.31%
SIC8. Professional Services		0.08%	0.87%	1.57%	0.24%	0.55%	0.08%	1.57%	2.13%		7.09%
SIC9. Public Administration											
Total	0.24%	5.20%	10.24%	29.06%	6.54%	11.65%	0.55%	28.35%	8.11%	0.08%	100.00%

Panel D. 2001-2022: 1,901 moves (Mean = 1.41%, HHI = 6.61%, Across-industry move = 45.13%)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	SIC0	SIC1	SIC2	SIC3	SIC4	SIC5	SIC6	SIC7	SIC8	SIC9	Total
SIC0. Agriculture, Forestry, Fishing			0.05%								0.05%
SIC1. Mining & Construction	0.05%	3.52%	0.47%	0.89%	0.16%	0.26%	0.11%	0.26%	0.11%	0.05%	5.89%
SIC2. Light Manufacturing		0.32%	13.52%	3.63%	0.74%	1.16%	0.26%	0.68%	0.63%	0.05%	20.99%
SIC3. Heavy Manufacturing	0.11%	0.79%	4.26%	16.25%	1.05%	1.63%	0.32%	3.79%	1.16%		29.35%
SIC4. Transportation & Public Utilities		0.42%	0.37%	1.21%	2.84%	0.53%	0.16%	1.37%	0.32%		7.21%
SIC5. Wholesale & Retail Trade		0.32%	0.74%	1.58%	0.37%	4.52%	0.47%	0.89%	0.21%		9.10%
SIC6. Finance, Insurance, Real Estate		0.16%	0.26%	0.32%	0.16%	0.26%	5.94%	0.89%	0.26%		8.26%
SIC7. General Services	0.05%	0.21%	1.37%	3.37%	0.68%	1.32%	0.58%	7.31%	0.79%		15.68%
SIC8. Professional Services		0.05%	0.63%	0.79%	0.05%	0.16%	0.26%	0.53%	0.95%		3.42%
SIC9. Public Administration			0.05%								0.05%
Total	0.21%	5.79%	21.73%	28.04%	6.05%	9.84%	8.10%	15.73%	4.42%	0.11%	100.00%

Table 5. Explaining the Trend in Executive Mobility

This table presents time-series regression estimates for the determinants of aggregate executive mobility from 1921 through 2011. The dependent variable is the number of CEO moves in year t divided by the number of CEOs in year $t-1$. A move is defined as a CEO who worked at a firm in year $t-2$ or $t-1$ moves to another firm in year t . Number of public firms is the total number of firms listed on the NYSE, Amex, and Nasdaq. Stddev. of ROA is the standard deviation of ROA of NYSE and Amex listed firms. $1/HHI$ across industries is the average inverse of the Herfindahl-Hirschman Index of the fraction of CEO moves across one-digit SIC industries over the past ten years. Number of New MBAs / New Engineering Masters is the number of business master's degree programs enrollment over the number of engineering master's degree programs enrollment in the US. Number of M&As is the number of completed M&A deals among US acquirors and targets. Avg. Option + Stock Granted / Total Compensation is the average fraction of option and restricted stock grants in total CEO compensation (= salary + bonus + option and restricted stock grants). % of Across-state Moves is the average proportion of across-state CEO moves over the past ten years. Real GDP growth is the growth rate of real GDP in the US. Credit spread (BAA - AAA) is the average yield difference of BAA and AAA rated 30-year US corporate bonds. Productivity is nominal GDP in year t over total private non-residential fixed assets in year $t-1$. 3-month T-bill (%) is the 3-month US Treasury bill rate. NBER Recession is the share of months in a year that are in recession, as classified by the NBER. σ (ROA) is the standard deviation of Compustat-firm ROA. Standard errors are displayed below coefficients with Newey-West (1987) adjustment for four lags.

	(1)	(2)	(3) [†]	(4) [†]	(5)	(6)
Dep. Var.	# CEO Move / # CEO					
Indep. Var.	Log # Firm	IPOs to Firms	Log # M&A	M&A to Firms	Inv HHI (SIC1)	MBA/STEM
Time period	1921-2022	1978-2022	1921-2022	1921-2022	1925-2022	1972-2022
Average effect	0.898*** (0.153)	0.146** (0.061)	0.238*** (0.053)	0.190*** (0.053)	0.754*** (0.172)	0.377*** (0.042)
<i>Era-specific effects</i>						
1920–1945	0.313 (0.198)		0.137* (0.078)	0.076* (0.043)	2.405*** (0.482)	
1946–1985	-0.092 (0.099)	-0.578** (0.227)	0.065* (0.036)	0.068*** (0.023)	0.230 (0.344)	-0.058 (0.043)
1986–2000	3.707*** (0.508)	0.485*** (0.089)	0.632*** (0.083)	0.792*** (0.062)	0.820*** (0.128)	1.420*** (0.137)
2001–2022	1.448*** (0.418)	0.178*** (0.052)	0.401** (0.203)	0.132 (0.237)	0.088 (0.074)	-0.057 (0.050)

Macroeconomic variables

$\sigma(\text{ROA})$	0.048 (0.065)	0.102 (0.067)	0.065* (0.038)	0.072 (0.044)	0.021 (0.062)	0.025 (0.054)
NBER Recession	0.088*** (0.029)	0.036 (0.101)	0.044* (0.023)	0.044* (0.023)	0.075* (0.039)	0.080 (0.078)
GDP growth	0.010 (0.022)	0.010 (0.053)	0.027** (0.013)	0.025* (0.013)	0.076*** (0.027)	0.013 (0.040)
Credit spread	-0.152*** (0.042)	-0.096 (0.087)	-0.078*** (0.022)	-0.063*** (0.024)	-0.096** (0.037)	-0.082 (0.058)
Productivity	-0.173*** (0.032)	0.059 (0.047)	-0.063*** (0.018)	-0.027 (0.024)	-0.023 (0.042)	0.150* (0.085)
3-month T-bill	0.144*** (0.054)	-0.040 (0.081)	0.038 (0.031)	0.063** (0.027)	0.103*** (0.036)	0.068 (0.053)
Era dummies	Y	Y	Y	Y	Y	Y
N	102	45	102	102	98	51
R^2	0.642	0.686	0.618	0.629	0.645	0.752

†: CEO moves exclude M&A or subsidiary-related events

Table 6. Executive Mobility and Redeployability of the Factors of Production: Time Series Regression Analysis

This table undertakes similar time series regression analysis to Table 5 for our factors of production redeployability measures. In column 1, we consider asset redeployability (1921-2022) and in column 2, we consider Labor redeployability (1976-2022). Controls are the same as in Table 5.

	(1)	(2)
Dep. Var.	# CEO Move / # CEO	
Indep. Var.	Asset Redeployability	Labor Redeployability
Time period	1921- 2022	1976- 2022
Average effect	0.188** (0.078)	0.182** (0.074)
<i>Era-specific effects</i>		
1920–1945	-0.181 (0.199)	
1946–1985	-0.276*** (0.085)	-1.071*** (0.269)
1986–2000	1.324*** (0.228)	0.876*** (0.137)
2001–2022	0.674*** (0.101)	0.279*** (0.069)
Macro controls	Y	Y
Era dummies	Y	Y
N	102	47
R^2	0.632	0.806

Table 7. Executive Mobility and Compensation: Time Series Regression Analysis

This table displays time series regressions mirroring the analysis in Figure 5.

	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var.	# CEO Move / # CEO					
Indep. Var.	Log Total Pay (Avg.)	Equity Share	Option Share	Pay-Size Ratio	Pay-Size Elasticity	Pay Dispersion
Time period	1936- 2022	1936- 2022	1936- 2022	1936- 2022	1938- 2022	1938- 2022
Average effect	-0.134 (0.133)	0.175 (0.175)	0.106* (0.061)	0.300*** (0.084)	0.261 (0.255)	-0.050 (0.208)
<i>Era-specific effects</i>						
1920–1945	-0.338 (0.483)	-0.301 (0.363)	-0.278* (0.164)	0.003 (0.112)	-0.169 (0.380)	-0.582 (0.957)
1946–1985	-0.183 (0.152)	-0.043 (0.208)	-0.037 (0.130)	-0.077 (0.060)	0.074 (0.551)	-0.387 (0.358)
1986–2000	-0.040 (0.234)	0.762*** (0.059)	0.543*** (0.044)	0.889*** (0.164)	0.496*** (0.062)	0.495*** (0.031)
2001–2022	-0.015 (0.142)	0.389 (0.449)	0.245*** (0.035)	0.718*** (0.252)	0.596** (0.287)	0.385*** (0.109)
Macro controls	Y	Y	Y	Y	Y	Y
Era dummies	Y	Y	Y	Y	Y	Y
N	87	87	87	87	85	85
R^2	0.421	0.655	0.693	0.673	0.673	0.720

Table 8. Cross-Sectional Determinants of Executive Mobility

This table presents logit regression estimation results for the determinants of executive mobility in the cross-section of firms from 1921 through 2011. 1(Move) is an indicator that is equal to one in year t if a CEO moves to another firm by year $t + 1$ or $t + 2$, and zero otherwise. A move is defined as a CEO who worked at a firm in year $t - 2$ or $t - 1$ moves to another firm in year t . 1 [Turnover] is an indicator that is equal to one in year t if a CEO leaves a firm by year $t + 1$ or $t + 2$, and zero otherwise. Average Industry ROA is the annual average of firm-level ROAs at the two-digit SIC industry level. Stddev of Industry ROA is the annual standard deviation of firm-level ROAs at the two-digit SIC industry level. All firm characteristics are defined in the Appendix. Columns 1-3 and 7-9 are on the full sample, and columns 4-6 are on the sample in which there is CEO turnover in year $t + 1$ or year $t + 2$ (1 [Turnover] = 1). ROA (before calculating industry average and dispersion) and Tobin's Q are Winsorized at the 1% and 99% levels, and all variables are standardized to unit variance in the regression. Standard errors are displayed below coefficients in parentheses and are double-clustered at the two-digit SIC industry and year levels.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	1 [Move] (Full sample)			1 [Move] (Turnover sample)			1 [Turnover] (Full sample)		
ROA	-0.194*** (0.034)	-0.199*** (0.033)	-0.200*** (0.034)	-0.290* (0.155)	-0.362** (0.152)	-0.292* (0.155)	-2.678*** (0.115)	-2.519*** (0.112)	-2.716*** (0.115)
Tobin's Q	-0.062** (0.028)	-0.074*** (0.028)	-0.074*** (0.028)	0.106 (0.148)	0.083 (0.149)	0.099 (0.149)	-1.969*** (0.099)	-2.009*** (0.101)	-2.051*** (0.101)
Log Assets	0.145*** (0.028)	0.152*** (0.028)	0.152*** (0.028)	0.362** (0.144)	0.398*** (0.145)	0.368** (0.146)	1.213*** (0.099)	1.213*** (0.099)	1.268*** (0.100)
CEO tenure	-0.337*** (0.019)	-0.337*** (0.019)	-0.337*** (0.019)	-0.197 (0.189)	-0.205 (0.189)	-0.198 (0.189)	-8.179*** (0.131)	-8.172*** (0.130)	-8.179*** (0.131)
Average Industry ROA	-0.119*** (0.037)		0.006 (0.063)	-0.812*** (0.185)		-0.721** (0.330)	0.761*** (0.133)		1.642*** (0.226)
σ (Industry ROA)		0.174*** (0.040)	0.179*** (0.068)		0.848*** (0.200)	0.130 (0.356)		-0.323** (0.152)	1.264*** (0.256)
Observations	181,356	181,356	181,356	25,691	25,691	25,691	181,356	181,356	181,356
Pseudo R-squared	0.004	0.004	0.004	0.009	0.009	0.009	0.070	0.070	0.070
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dep. var.	0.830%			3.826%			14.166%		